

"Smart Library Database Engineering and Student Services Portal"

A Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

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In

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

&

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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Declaration

The Project Report entitled **“Secure and Scalable Healthcare Appointment Scheduling Through Distributed Backend Development”** is a record of Bonafide work of **BRUNDAVANAM HARINI – 2520030384, BHAVIKA PAWAR – 2520030616, 2520030420 – HITESH REDDY** submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering (or) Computer Science and Information Technology to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is certify that the mini project based report entitled **“Secure and Scalable Healthcare Appointment Scheduling Through Distributed Backend Development”** is a bonafide work done and submitted by **BRUNDAVANAM HARINI – 2520030384, BHAVIKA PAWAR – 2520030616, 2520030420 – HITESH REDDY** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science Engineering, K L (Deemed to be University), during the academic year **2025-2026**.

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Abstract

The Smart Library Database System and Student Services Portal is a full-stack web application designed to simplify library management and provide convenient digital services for students. Students can register and log in, search and filter books, view book details and availability, borrow and return books, and track their borrowing history.

The system also provides an admin portal where librarians can add, update, delete, and manage books, students, categories, and borrowing record. The application uses React.js for the frontend, Node.js and Express.js for backend development, and MySQL for database management.

The frontend communicates with the backend through APIs, while the database securely stores user, book, and transaction information. The system reduces manual record keeping, improves data accuracy, and makes library services faster and more accessible.

Overall, the project demonstrates practical concepts of full-stack web development, database management, authentication, CRUD operations, API communication, and system integration.

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CHAPTER -1 INTRODUCTION

1.1 Background of the Project

Libraries play an important role in educational institutions by providing students and faculty with access to books, academic resources, and other learning materials. Traditionally, many library activities such as maintaining book records, registering members, issuing books, recording returns, managing reservations, and tracking fines have been performed manually or through basic library management systems. Although these approaches can support basic operations, they may become time-consuming and difficult to maintain when the number of books, students, and transactions increases.

The growth of web technologies has created opportunities to improve traditional library services through centralized digital platforms. A web-based library management system can provide users with faster access to library information while reducing the amount of manual record keeping required by librarians. Students can search for books, check availability, manage borrowing activities, and view their previous library transactions without depending entirely on the physical library counter.

The proposed project, “**Smart Library Database System and Student Services Portal,**” is a full-stack web application designed to simplify library management and provide convenient digital services to students. The system provides a centralized platform through which students can register and log in, search and filter books, view book details and availability, borrow and return books, reserve unavailable books, view fines, and track their borrowing history.

The system also provides an administrative portal for librarians. Administrators can manage books, students, categories, borrowing records, reservations, and other library-related information. This provides centralized control over the library database and helps maintain accurate and organized records.

The project is developed using **React.js** for the frontend, **Node.js and Express.js** for backend development, and **MySQL** for database management. The frontend communicates with the backend through APIs, while the MySQL database stores structured information related to members, books, borrowing transactions, reservations, fines, and other library activities.

An additional feature of the proposed system is offline accessibility through caching. Previously accessed information can be made available even when the internet connection is temporarily unavailable. Background synchronization can update the system when the connection is restored. Notifications can also be used to inform users about due dates, reservations, fines, and other important library updates.

The project combines several important areas of computer science, including:

1. Full-stack web application development
2. Database management
3. Relational database design
4. REST API development
5. Authentication and authorization
6. CRUD operations
7. SQL queries and database transactions
8. Offline caching and synchronization
9. Software engineering and modular design
10. User interface and user experience design

The overall purpose of the project is to provide a reliable, organized, and convenient digital library platform that improves access to library services while reducing manual administrative work.

.2 Problem Statement

Traditional library management processes can create difficulties for both students and librarians. Students may need to visit the library physically or depend on library staff to check book availability, borrowing information, due dates, reservations, and fines. This can result in unnecessary waiting and inconvenience.

Maintaining library records manually can also introduce errors and duplication. Librarians need to maintain information about students, books, categories, borrowing transactions, returns, reservations, and fines. As the number of users and books increases, maintaining these records becomes more difficult.

Another problem is limited accessibility when the internet connection is unavailable. Basic web-based systems generally depend on continuous connectivity, which can prevent users from accessing previously available information during temporary network interruptions.

Students may also fail to receive timely information about due dates, reservations, or fines if the system does not provide an effective notification mechanism. This can result in delayed returns and additional administrative work.

The absence of a centralized system can therefore create problems such as:

1. Difficulty in maintaining accurate library records.
2. Time-consuming manual book and member management.
3. Difficulty checking book availability quickly.
4. Limited access to borrowing history.
5. Difficulty managing reservations.
6. Difficulty tracking fines.
7. Limited accessibility during temporary internet interruptions.
8. Lack of timely notifications and reminders.
9. Increased workload for librarians.
10. Difficulty generating useful library reports.

Therefore, there is a need for a centralized digital library system that allows students to conveniently access library services while providing librarians with efficient tools for managing library records.

The proposed Smart Library Database System addresses these problems by providing a centralized web application with student services, administrative management, database-driven operations, offline caching, background synchronization, and notifications.

1.3 Scope of the Project

The scope of the Smart Library Database System and Student Services Portal covers the development of a web-based platform for managing library resources and providing digital services to students.

The primary scope of the project includes student registration and authentication. Students can create accounts, log in to the system, and manage their profiles. Authentication helps ensure that library-related information is accessible only to authorized users.

The project also covers book and asset management. Library administrators can maintain book information including title, author, category, and availability. Books can be added, updated, deleted, searched, and organized according to different categories.

Borrowing management is another major part of the system. Students can borrow available books and return them through the system. Borrowing records maintain information about the transaction and help track the student's library activity.

The system also supports reservation management. If a required book is unavailable, students can reserve the book. Reservation records maintain the relationship between the student and the selected book.

Fine management is included to maintain fine-related information associated with library

activities. Fines can be tracked and managed using database operations.

The Search and Reports module allows users to search for books and retrieve useful library information. Administrators can use database queries to generate reports based on library records.

The Member History module maintains information about previous library activities, including borrowed and returned books. This allows students and administrators to retrieve historical information efficiently.

The Admin Management module provides librarians with centralized control over students, books, categories, borrowing records, reservations, and other library-related information.

The project also considers offline caching and background synchronization. Previously accessed information can remain available during temporary network interruptions, while synchronization can update information after connectivity is restored.

The technical scope includes:

- Responsive web interface
- Student registration and login
- Authentication and authorization
- Book searching and filtering
- Book availability checking
- Borrowing and returning
- Reservation management
- Fine management
- Member borrowing history
- Administrative management
- REST API communication
- MySQL database management
- Offline caching
- Background synchronization
- Notifications
- API testing using Postman
- Version control using Git and GitHub

The current project primarily focuses on library management and student services. Advanced features such as online fine payment, barcode or QR-code scanning, AI-based book recommendations, e-book access, and digital archive management can be considered for future development.

1.4 Objectives of the Project

The main objective of the Smart Library Database System and Student Services Portal is to develop a user-friendly and centralized digital platform that simplifies library management and provides convenient services to students.

The major objectives of the project are:

1. To develop a centralized web portal for managing library resources and student services.
2. To provide students with secure registration and login functionality.
3. To allow students to search and filter books easily.
4. To provide information about book details and availability.
5. To enable students to borrow and return books through the system.
6. To provide reservation functionality for unavailable books.
7. To maintain and display student borrowing history.
8. To manage fine-related information associated with library activities.

9. To provide librarians with an administrative portal for managing library records.
10. To maintain student, book, category, borrowing, reservation, and fine information in a structured MySQL database.
11. To reduce manual record keeping and improve the accuracy of library information.
12. To implement CRUD operations for important library entities.
13. To use database constraints and relationships to maintain data consistency.
14. To provide search and reporting functionality using SQL queries.
15. To provide offline access to previously cached information.
16. To implement background synchronization when network connectivity is restored.
17. To provide notifications for important library updates such as due dates and reservations.
18. To demonstrate full-stack development using React.js, Node.js, Express.js, and MySQL.
19. To provide REST API communication between the frontend and backend.
20. To create a modular and maintainable architecture that can be extended in the future..

1.5 Importance of the Application

The Smart Library Database System is important because it improves the accessibility and efficiency of library services within an educational institution.

For students, the system provides a convenient way to search for books and check their availability without depending completely on the physical library counter. Students can also manage borrowing activities, reservations, fines, and borrowing history through a single platform.

For librarians, the system reduces the amount of manual work involved in maintaining library records. Instead of maintaining separate records for books, students, and transactions, the information can be managed through a centralized database.

The system also improves data organization. Since information is stored in a structured relational database, relationships between students, books, borrowing records, reservations, and fines can be maintained more effectively.

The notification feature can improve communication by informing students about important library events such as upcoming due dates, reservations, and other updates.

Offline caching provides additional convenience because previously accessed information can remain available during temporary network interruptions. Background synchronization can help update information when connectivity becomes available again.

The application is also important from a software engineering perspective because it demonstrates the practical implementation of:

- Frontend development
- Backend development
- Database management
- REST API communication
- Authentication
- CRUD operations
- SQL queries
- Transactions
- Database constraints
- Modular system design
- Offline caching
- Synchronization
- Notification functionality

Overall, the application can:

- Reduce manual library work.
- Improve data accuracy.

- Save students' time.
- Improve book search and availability checking.
- Simplify borrowing and returning.
- Improve reservation management.
- Provide better access to borrowing history.
- Improve fine tracking.
- Provide centralized administration.
- Improve communication through notifications.
- Provide limited offline accessibility.
- Support future expansion of library services.

1.6 Target Users / Audience

The primary target users of the system are **patients** who need to schedule and manage healthcare. The primary users of the system are students who use library resources and services.

The second major group of users is librarians or administrators who are responsible for maintaining library records and managing library operations.

The target users include:

1.6.1 Students

Students can:

- Register for an account.
- Log in securely.
- Search for books.
- Filter books.
- View book details.
- Check book availability.
- Borrow books.
- Return books.
- Reserve unavailable books.
- View borrowing history.
- View fine-related information.
- Receive library notifications.
- Manage their profile.

1.6.2 Librarians / Administrators

Administrators can:

- Manage student records.
- Add books.
- Update book information.
- Delete book records.
- Manage book categories.
- Monitor borrowing transactions.
- Manage reservations.
- Manage fine-related records.

- View library reports.
- Maintain database information.
- Manage overall library operations.

1.6.3 Educational Institutions

Educational institutions can use the system to improve the efficiency of their library operations, reduce manual record keeping, and provide students with convenient digital access to library services.

CHAPTER -2 SYSTEM REQUIREMENTS

2.1 Hardware Requirements

Hardware requirements define the physical resources required to develop, test, and operate the Smart Library Database System and Student Services Portal.

Since the proposed system is a web-based application, specialized hardware is not required. A standard computer with sufficient processing power, memory, storage, and network connectivity is adequate for development and testing.

2.1.1 Development Computer

The development computer is used by the project team to write source code, run the React frontend, execute the Node.js and Express.js backend, manage the MySQL database, test APIs, and access the application through a web browser.

Recommended configuration:

A recommended configuration is:

Hardware Component	Minimum Requirement	Recommended Requirement
Processor	Intel Core i3 / equivalent	Intel Core i5 or higher
RAM	8 GB	16 GB
Storage	256 GB	512 GB SSD
Display	1366 × 768	1920 × 1080
Network	Stable Internet	High-speed Internet
Keyboard	Standard keyboard	Standard keyboard
Mouse	Standard mouse	Standard mouse

A multi-core processor is useful because several development applications may be running simultaneously, such as Visual Studio Code, Node.js, MySQL, MySQL Workbench, Postman, and a web browser.

At least 8 GB RAM is suitable for basic development, while 16 GB provides better performance when multiple applications are used simultaneously. SSD storage is preferred because it provides faster file access and improves application startup and database operations.

2.1.2 Storage Requirements

Storage is required for:

- Project source code
- Node.js packages and dependencies
- MySQL database files
- API testing data
- Documentation
- Screenshots
- Git repositories
- Backup files
- Configuration files

A minimum of 256 GB storage is sufficient for development, while 512 GB SSD storage is recommended for smoother operation and future expansion.

2.1.3 Internet Connectivity

A stable internet connection is useful for:

- Installing Node.js packages and dependencies.
- Accessing technical documentation.
- Synchronizing code with GitHub.
- Testing APIs.
- Downloading development tools.
- Collaborating with team members.
- Future deployment of the application.

The application also considers offline caching so that previously accessed information can remain accessible during temporary network interruptions.

2.1.4 Client Hardware Requirements

Students and administrators do not require high-end hardware because the application is accessed through a web browser.

The system can be accessed using:

- Desktop computers
- Laptops
- Tablets
- Smartphones

A basic device with approximately 4 GB RAM, a modern web browser, and a stable internet connection should be sufficient for normal usage.

2.1.5 Server Hardware Requirements

During development, the Node.js backend and MySQL database can run on the development computer.

For larger-scale deployment, the application can be hosted on a dedicated or cloud server.

A basic server configuration may include:

- Multi-core processor
- 8–16 GB RAM
- SSD storage
- Stable network connection
- Appropriate operating system
- Database storage
- Backup facilities

Server resources can be increased depending on the number of students, books, and transactions handled by the system.

2.2 Software Requirements

Software requirements define the software environment needed to develop, test, run, and maintain the Smart Library Database System.

2.2.1 Operating System

The application can be developed and tested using:

- Windows 10
- Windows 11
- Ubuntu Linux
- macOS

Windows provides convenient access to Visual Studio Code, Node.js, MySQL, MySQL Workbench, browsers, and Postman.

Linux can also be used for development and server deployment.

2.2.2 Frontend Software

The frontend is developed using React.js.

React.js is responsible for creating the user interface of the application. It can be used to develop:

- Login pages
- Registration pages
- Student dashboard
- Book search interface
- Book details
- Borrowing pages
- Reservation pages
- Fine information
- Borrowing history
- Admin dashboard

The frontend communicates with the backend through REST APIs.

2.2.3 Backend Software

The backend is developed using Node.js and Express.js.

Node.js provides the runtime environment for executing server-side JavaScript, while Express.js provides a framework for creating REST APIs.

The backend is responsible for:

- User registration
 - User login
 - Authentication
 - Book management
 - Student management
 - Category management
 - Borrowing operations
 - Returning books
 - Reservation management
 - Fine management
 - Search operations
 - Reports
 - Administrative operations
 - Notifications
-

2.2.4 Database Software

MySQL is used as the relational database management system.

The database stores important information including:

- Student/member details
- Book details
- Category details
- Borrowing records
- Return information
- Reservation records
- Fine information
- Administrative information
- Library transaction data

MySQL provides structured data storage and allows relationships to be maintained between different entities.

2.2.5 Authentication Software

MySQL Workbench is used for:

- Creating databases.
- Creating tables.
- Defining relationships.
- Writing SQL queries.
- Managing constraints.
- Viewing database structures.
- Executing database operations.
- Inspecting records.
- Testing database functionality.

2.2.6 API Testing Software

Postman is used for testing REST APIs developed using Node.js and Express.js.

HTTP methods such as:

GET

POST

PUT

DELETE

can be tested using Postman.

Examples of APIs that can be tested include:

Registration API

Login API

Book API

Member API

Borrowing API

Return API

Reservation API

Fine API

Search API

Admin API

Testing APIs independently helps identify backend problems before integrating the APIs with the frontend.

2.2.7 Web Browser

A modern web browser is required to access and test the application.

Supported browsers can include:

Google Chrome

Microsoft Edge

Mozilla Firefox

The browser is used to:

Access the application.

Test forms.

Check responsive design.

Verify frontend functionality.

Inspect API responses.

Debug frontend issues.

2.3 Development Tools and Frameworks

2.3.1 Visual Studio Code

Visual Studio Code is used as the primary code editor for the project.

It provides an environment for working with:

JavaScript

React.js

Node.js

Express.js

SQL

HTML

CSS

It also provides useful features such as syntax highlighting, debugging, extensions, integrated terminal support, and project management.

2.3.2 Node.js

Node.js is used as the runtime environment for the backend.

It allows JavaScript code to execute on the server and supports the development of backend services and REST APIs.

2.3.3 Express.js

Express.js is used as the backend framework.

It simplifies the creation of REST APIs and allows the application to handle HTTP requests and responses.

The backend can be divided into different components such as:

Authentication APIs

Member APIs

Book APIs

Borrowing APIs

Reservation APIs

Fine APIs

Search and Report APIs

Admin APIs

2.3.4 React.js

React.js is used to develop the frontend interface.

Its component-based structure allows different parts of the application to be developed and reused independently.

Examples include:

Navigation components

Book cards

Search components

Forms

Student dashboard

Admin dashboard

Borrowing components

Reservation components

2.3.5 MySQL

MySQL provides the relational database layer.

It supports:

Structured data storage

SQL queries

Table relationships

Primary keys

Foreign keys
Constraints
Data retrieval
Data modification
Transactions
Database administration

2.3.6 Git

Git is used for version control.
It allows the development team to:
Track code changes.
Create branches.
Commit changes.
Merge changes.
Restore earlier versions.
Maintain project history.

2.3.7 GitHub

GitHub is used as the remote repository for storing and sharing project source code.
It supports:
Team collaboration
Source-code backup
Version history
Repository management

2.3.8 Postman

Postman is used to test and verify REST API functionality.
It allows the team to send requests to the backend and examine the returned responses before frontend integration.

2.4 Summary of System Requirements

Category	Technologies / Requirements
Operating System	Windows / Linux / macOS
Frontend	React.js
Backend	Node.js, Express.js
Database	MySQL
Database Tool	MySQL Workbench
API Testing	Postman
Code Editor	Visual Studio Code
Version Control	Git
Repository	GitHub
Browser	Chrome / Edge / Firefox
RAM	8 GB minimum, 16 GB recommended
Storage	256 GB minimum, SSD recommended
Internet	Stable Internet connection

2.5 Overall Requirement Conclusion

The proposed Smart Library Database System requires moderate hardware and commonly available software tools. The combination of React.js, Node.js, Express.js, MySQL, MySQL Workbench, Postman, Visual Studio Code, Git, and GitHub provides a complete development and testing environment. This technology selection supports the project's goal of creating a centralized, user-friendly, and database-

driven library management system. The architecture also provides a foundation for implementing offline caching, synchronization, notifications, and future extensions.

CHAPTER 3 – TECHNOLOGY STACK

The technology stack defines the programming languages, frameworks, database technologies, development tools, API testing tools, and version-control systems used to build the Smart Library Database System and Student Services Portal.

The proposed system uses React.js for the frontend, Node.js and Express.js for backend development, MySQL for database management, Postman for API testing, and Git and GitHub for version control and collaboration. These technologies provide a practical foundation for developing a full-stack library management application.

3.1 Front-End Design – React.js

The frontend represents the presentation layer of the Smart Library Database System. It is the part of the application through which students and administrators interact with the system.

React.js is used to develop a responsive and component-based user interface.

The major responsibilities of the frontend include:

- Providing registration and login pages.
- Displaying student information.
- Providing book search functionality.
- Displaying book details.
- Showing book availability.
- Providing borrowing functionality.
- Providing return functionality.
- Providing reservation functionality.
- Displaying fine information.
- Displaying borrowing history.
- Providing administrative interfaces.
- Displaying notifications.

The React frontend communicates with the Node.js and Express.js backend through REST APIs.

When a user performs an action, the frontend sends an API request to the backend. The backend processes the request and returns a response. React then updates the user interface according to the response.

3.1.1 React Components

React uses a component-based architecture.

The application can be organized into reusable components such as:

- Header
- Navigation bar

- Login form
- Registration form
- Book card
- Book search
- Book details
- Borrowing form
- Reservation form
- Fine display
- History display
- Student dashboard
- Admin dashboard
- Notification component

The component-based approach improves maintainability because individual interface elements can be developed and modified independently.

3.1.2 User Interface

The user interface is designed to provide simple navigation between different library services.

Students can access functions such as:

- Search Books
- View Availability
- Borrow Book
- Return Book
- Reserve Book
- View History
- View Fines
- View Notifications

Administrators can access functions such as:

- Manage Students
- Manage Books
- Manage Categories
- Manage Borrowing
- Manage Reservations
- Manage Fines
- View Reports

3.2 Back-End – Node.js and Express.js

The backend is the main processing layer of the application.

Node.js provides the runtime environment, while Express.js provides the framework for creating REST APIs.

The backend is responsible for implementing the main functionality of the system.

These functions include:

1. User registration
2. User login
3. Authentication
4. Member management
5. Book management
6. Category management
7. Borrowing management
8. Return management
9. Reservation management
10. Fine management
11. Search functionality
12. Report generation
13. Notification-related operations
14. Administrative operations

3.2.1 Node.js

Node.js is used to execute JavaScript on the server side.

It provides the environment required to implement backend operations such as:

- Processing user requests.
- Handling API requests.
- Communicating with MySQL.
- Managing library transactions.
- Performing CRUD operations.
- Processing authentication.
- Returning API responses.

Node.js is suitable for this project because the frontend and backend can both use JavaScript, simplifying development and integration.

3.2.2 Express.js

Express.js is used to create the REST API layer.

It receives HTTP requests from the React frontend and sends appropriate responses.

The backend can be divided into multiple API groups:

Express.js Backend

├──	Authentication	APIs
├──	Member	APIs
├──	Book	APIs
├──	Borrowing	APIs
├──	Reservation	APIs
├──	Fine	APIs
├──	Search and Report	APIs
└──	Admin APIs	

This modular structure makes the backend easier to develop, test, maintain, and extend.

3.3 Database – MySQL

MySQL is used as the relational database management system for the project.

The database stores and manages the information required for library operations.

The main information stored includes:

- Member information
- Book information
- Category information
- Borrowing information
- Return information
- Reservation information
- Fine information
- Administrator information
- Notification-related information

A relational database is suitable for the project because library entities have clearly defined relationships.

3.3.1 Member Data

The member-related table can store information such as:

- Member ID
- Member name
- Email
- Contact information
- Login credentials
- Other required profile information

Each member can have multiple borrowing and reservation records.

3.3.2 Book Data

The book-related table can contain:

- Book ID
- Book title
- Author
- Category ID
- Availability status
- Other book-related details

Book records allow users to search for resources and determine whether a book is currently available.

3.3.3 Borrowing Data

The borrowing table is one of the important components of the database.

It can contain:

- Borrowing ID
- Member ID
- Book ID
- Borrow date
- Due date
- Return date
- Borrowing status

The borrowing information allows the system to maintain a complete record of book transactions.

3.3.4 Reservation Data

The reservation table maintains information about books reserved by members.

It can contain:

- Reservation ID
- Member ID
- Book ID
- Reservation date
- Reservation status

This allows the system to track reservations for unavailable books.

3.3.5 Fine Data

The fine table stores fine-related information.

It can contain:

- Fine ID
- Member ID
- Borrowing ID
- Fine amount
- Fine status
- Fine date

Fine records help administrators track fines associated with library transactions.

3.4 Authentication and Authorization

Authentication is used to verify the identity of users before allowing access to protected parts of the application.

The general process is:

User

↓

Enter

Login

Credentials

↓

Backend

↓

Verify

Credentials

↓

Authentication

Successful

↓

Access Protected Functions

Authorization determines what functions a user can access.

For example, a student should be able to access their own library services, while an administrator should have access to management functions.

The system therefore provides separate functionality for students and administrators.

3.5 REST API

REST APIs provide communication between the React frontend and the Node.js/Express.js backend.

The frontend sends HTTP requests to the backend, and the backend processes the request and returns a response.

Typical operations include:

Method	Purpose
--------	---------

GET Retrieve information
 POST Create new information
 PUT Update existing information
 DELETE Remove information

Possible API operations include:

POST /register
 POST /login
 GET /books
 GET /books/{id}
 POST /borrow
 PUT /borrow/{id}
 POST /reserve
 GET /history
 GET /fines
 POST /admin/books
 PUT /admin/books/{id}
 DELETE /admin/books/{id}

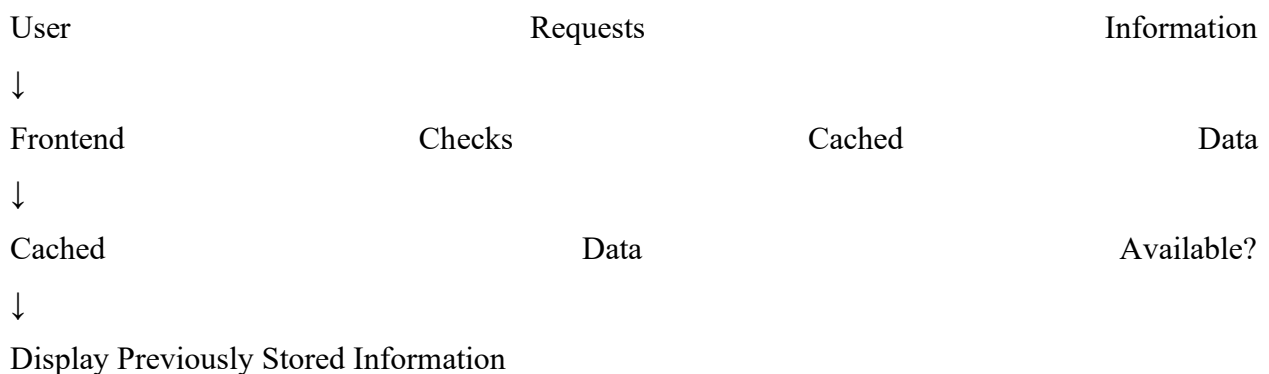
The use of REST APIs keeps the frontend and backend logically separate and makes the system easier to maintain.

3.6 Offline Caching and Background Synchronization

Offline caching is an important supporting feature of the proposed system.

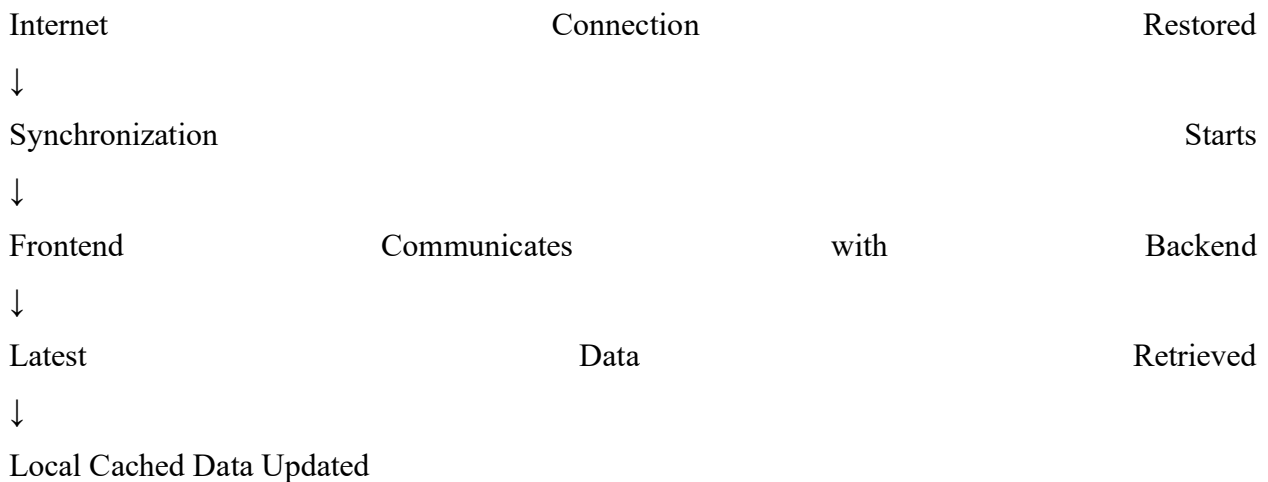
The system can store previously accessed information locally so that users can continue viewing selected information during temporary network interruptions.

The basic process is:



When network connectivity is restored, background synchronization can update the locally cached information with the latest information from the backend.

The general process is:



This approach improves accessibility and reduces dependence on continuous connectivity for previously accessed information.

3.7 Notifications

The notification functionality is designed to keep students informed about important library activities.

Notifications can be associated with:

- Upcoming due dates
- Book reservations
- Borrowing updates
- Return-related information
- Fine-related updates
- Other library announcements

Notifications improve communication between the library system and its users.

3.8 Version Control – Git and GitHub

Git is used for version control, while GitHub is used as the remote repository.

Git allows the development team to:

- Track changes.
- Create branches.
- Commit code.
- Merge changes.
- Restore earlier versions.
- Maintain development history.

GitHub provides a centralized location for storing and sharing project source code.

This is particularly useful because the project work is divided among three team members and

different modules.

3.9 API Testing – Postman

Postman is used to test REST APIs developed using Node.js and Express.js.

Before connecting the backend to the frontend, individual API endpoints can be tested using Postman.

For example:

Registration Testing

POST /register

↓

Send member information

↓

Backend

↓

Store information in MySQL

↓

Return response

Login Testing

POST /login

↓

Send credentials

↓

Verify credentials

↓

Return authentication response

Book Testing

GET /books

↓

Retrieve book information

↓

Return book records

Borrowing Testing

POST /borrow

↓

Send borrowing details

Check

book

↓

Store

borrowing

transaction

Return response

Postman helps identify API errors, incorrect responses, validation problems, and database-related issues before frontend integration.

3.10 Integrated Technology Stack

The complete technology stack can be represented as:

USERS



Offline

Caching

/

Synchronization

|



Notifications

Supporting tools include:

VS Code → Development

Git/GitHub → Version Control

Postman → API Testing

MySQL Workbench → Database Management

3.11 Summary

The technology stack provides the necessary components for developing the Smart Library Database System and Student Services Portal.

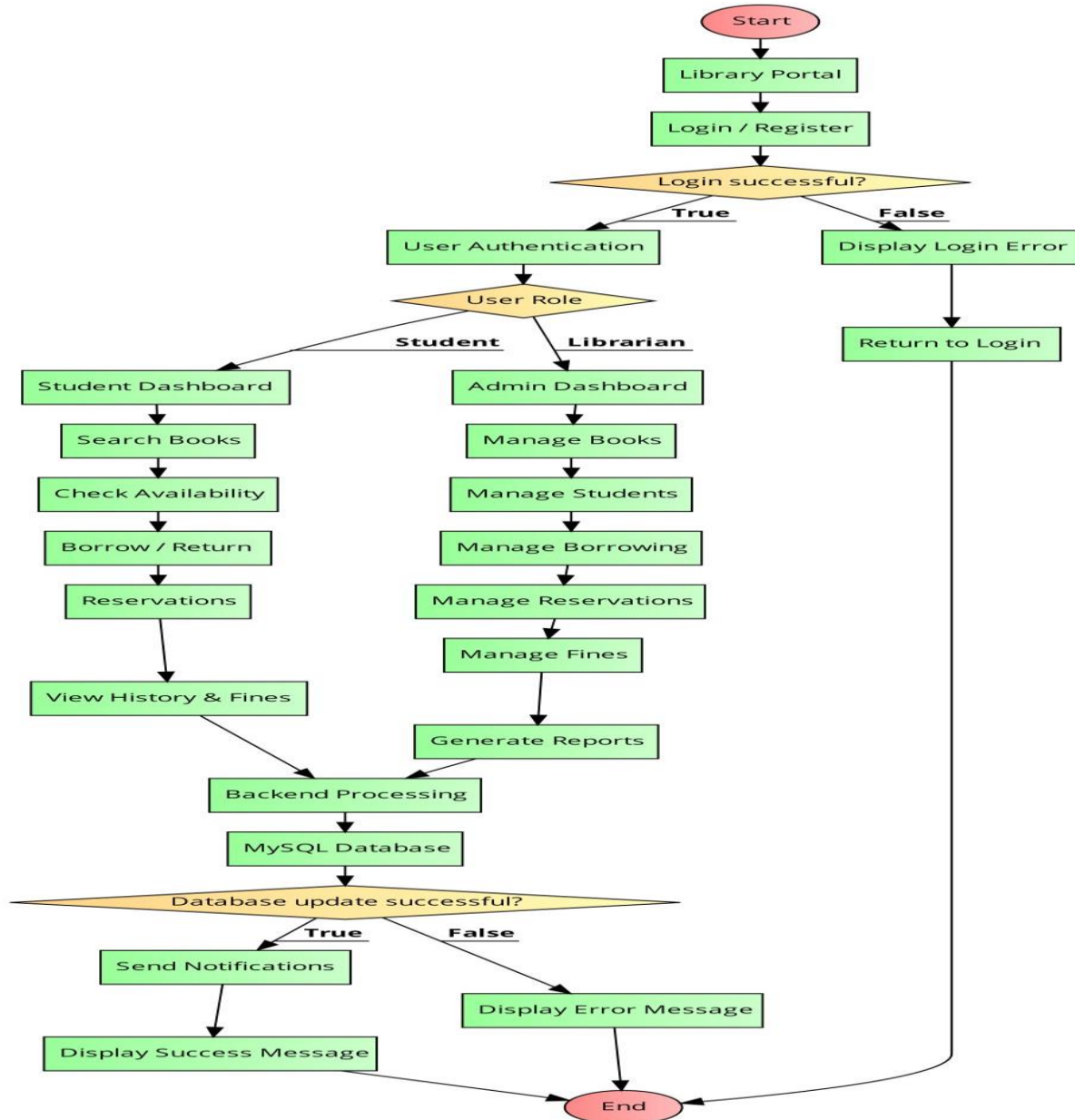
React.js provides the frontend interface, while Node.js and Express.js implement the backend and REST APIs. MySQL provides structured and relational data storage. Authentication and authorization protect user functionality, while offline caching and synchronization improve accessibility.

Git and GitHub support collaborative development and version control, while Postman is used to test backend APIs. MySQL Workbench supports database design and management.

Together, these technologies provide a practical foundation for developing a centralized, modular, maintainable, and scalable library management system.

CHAPTER -4 SYSTEM ARCHITECTURE

4.1 High-level architecture diagram



4.2 Users of the System

The system is designed primarily for two categories of users:

Students / Members

Librarians / Administrators

4.2.1 Student / Member

Students are the primary users of the library portal.

They can:

Register for an account.

Log into the system.
Search for books.
Filter books.
View book details.
Check availability.
Borrow books.
Return books.
Reserve unavailable books.
View borrowing history.
View fine information.
Receive notifications.
Manage their profile.
Students interact with the system through the React frontend, while all major database operations are performed through the backend.

4.2.2 Librarian / Administrator

The administrator is responsible for managing the overall library system.

The administrator can manage:

Student records
Books
Categories
Borrowing records
Reservations
Fine records
Library reports
Other library-related information

The administrator uses the system through a dedicated administrative interface.

4.3 Frontend Layer

The frontend is the presentation layer of the system and is developed using React.js.

It provides the interface through which students and administrators interact with the application.

The main frontend functions include:

Registration
Login
Book search
Book filtering
Book details
Availability checking
Borrowing
Returning
Reservation
Fine information
Borrowing history
Notifications
Admin dashboard

When a user performs an operation, the React frontend sends a request to the Express.js backend through a REST API.

The backend processes the request and returns the required information. The frontend then displays the result to the user.

4.4 Backend Layer

The backend is developed using Node.js and Express.js.

It acts as the central processing layer between the frontend and database.

The backend:

Receives API requests.
Validates input.
Performs business logic.
Communicates with MySQL.
Processes transactions.
Applies authorization rules.
Returns API responses.

The backend can be divided into different API components.

Authentication APIs

Handle:

Registration

Login

Authentication

Authorization

Member APIs

Handle:

Member information

Member profiles

Member records

Book APIs

Handle:

Book information

Book search

Book availability

Book creation

Book updates

Book deletion

Borrowing APIs

Handle:

Book borrowing

Book returning

Borrowing records

Due dates

Borrowing status

Reservation APIs

Handle:

Creating reservations

Viewing reservations

Updating reservation status

Fine APIs

Handle:

Fine information

Fine records

Fine status

Search and Report APIs

Handle:

Book searches

Library information

Reports

Aggregated information

Admin APIs

Handle administrative operations related to students, books, categories, borrowing, reservations, and fines.

4.5 Authentication and Authorization

Authentication is used to verify that a user is a registered member of the system.

The general process is:

User Login



Credentials Submitted



Backend Verification



Authentication Successful



Access to Protected Functions

Authorization determines the functions that a user is permitted to access.

For example:

User Type	Main Access
-----------	-------------

Student	Profile, book search, borrowing, reservations, history, fines
---------	---

Administrator	Student, book, category, borrowing, reservation, fine and report management
---------------	---

This separation helps prevent unauthorized access to administrative functions.

4.6 Database Layer

The database layer uses MySQL to store the information required by the application.

The major information stored includes:

Member details

Book details

Category details

Borrowing details

Reservation details

Fine details

Administrator information

Notification-related information

The backend communicates with MySQL whenever information needs to be created, retrieved, updated, or deleted.

A simplified relationship can be represented as:

Member ————— Borrowing ————— Book



Fine

The database relationships help maintain consistency between library records.

4.7 Book and Asset Management

The Book and Asset Management module is responsible for maintaining information about books and library assets.

Administrators can:

Add books.

Update books.

Delete books.

Search books.

Assign categories.

Manage availability.

Students can search for books and check their availability.

The general process is:

Search Book



Select Book

↓
View Book Details

↓
Check Availability

↓
Borrow or Reserve

This module is directly connected to the borrowing and reservation modules.

4.8 Borrowing Management

The Borrowing Management module handles the process of borrowing and returning books.

The borrowing process is:

Student Searches Book

↓
Select Book

↓
Check Availability

↓
Book Available?

↓
Yes

↓
Borrow Book

↓
Create Borrowing Record

↓
Update Book Availability

↓
Display Confirmation

When the book is returned, the system updates the borrowing record and the book's availability.

The borrowing process should maintain reliable database operations so that book availability and transaction records remain consistent.

4.9 Reservation Management

The Reservation Management module allows students to reserve books that are currently unavailable.

The process is:

Search Book

↓
Book Unavailable

↓
Select Reserve

↓
Create Reservation

↓
Store Reservation in MySQL

↓
Display Reservation Status

Reservation records maintain the relationship between the member and the selected book.

When the book becomes available, the system can use the reservation information to support the reservation process.

4.10 Fine Management

The Fine Management module handles fine-related information associated with library activities.

Fine information can be maintained for situations where library rules result in a fine.

The system can maintain:

Fine ID

Member ID
Borrowing ID
Fine amount
Fine status
Fine date

The fine information allows students and administrators to track outstanding library-related fines. Database operations, procedures, and triggers can be used as appropriate to maintain fine-related operations.

4.11 Search and Reports

The Search and Reports module provides functionality for retrieving useful information from the database.

Students can search for books based on available search and filtering options.

Administrators can use reports to retrieve information about:

Books
Students
Borrowing records
Reservations
Fines

Library activities

SQL techniques such as joins, subqueries, aggregate functions, and other database operations can be used to generate useful information.

4.12 Member History

The Member History module maintains records of a student's previous library activities.

The history can include:

Previously borrowed books
Returned books
Borrowing dates
Return dates
Transaction status

A simplified flow is:

Student
↓
Request History
↓
Backend
↓
MySQL
↓
Retrieve Borrowing Records
↓
Display History

Views, joins, and aggregate functions can be used to retrieve historical information efficiently.

4.13 Offline Caching and Background Synchronization

Offline caching improves accessibility during temporary network interruptions.

When information has previously been accessed, selected information can be stored locally.

The process can be represented as:

User Requests Data

↓
Internet Available?

└── Yes → Backend → MySQL → Latest Data

└─ No → Cached Data → Display Previously Accessed Information

When the connection becomes available again, background synchronization can occur:

Connection Restored



Synchronization Starts



Backend Request



Latest Information Retrieved



Cache Updated

This feature improves the usability of the application when network connectivity is temporarily unavailable.

4.14 Notification Module

The notification module helps provide important updates to students.

Notifications can relate to:

Due dates

Reservations

Borrowing updates

Return-related information

Fines

Other library announcements

The general process is:

Library Event



Backend Detects Event



Notification Generated



Notification Stored / Delivered



Student Views Notification

The notification system helps improve communication between students and the library.

4.15 API Testing and Development Tools

Postman is used to test backend REST APIs independently of the frontend.

It can be used to test:

Registration

Login

Member APIs

Book APIs

Borrowing APIs

Reservation APIs

Fine APIs

Search APIs

Admin APIs

Git is used to track source-code changes, while GitHub is used for repository management and team collaboration.

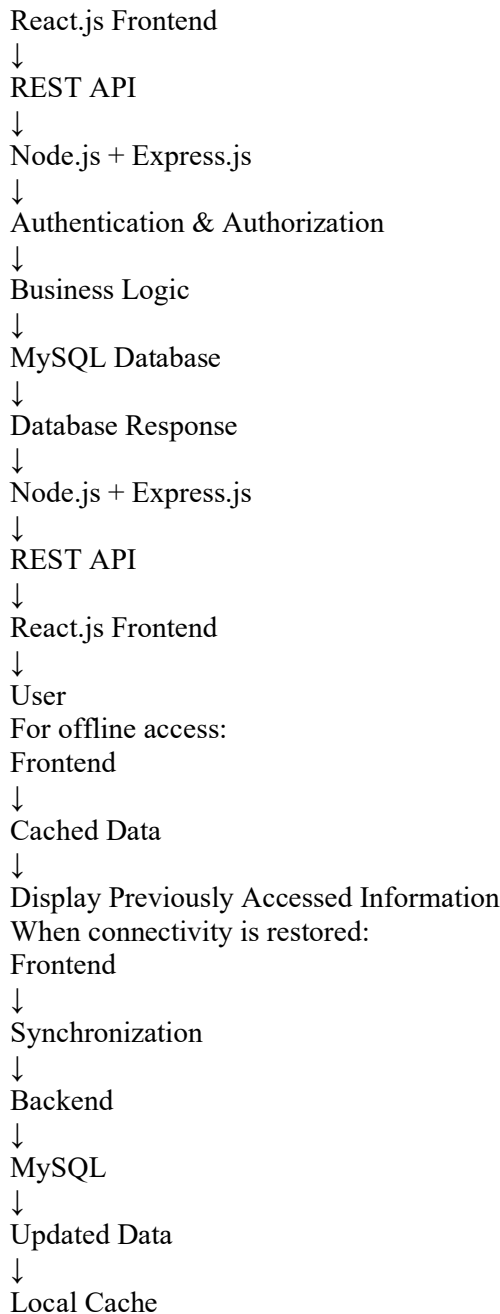
MySQL Workbench is used for database design, SQL queries, and database administration.

4.16 Overall Data Flow

The complete data flow of the system can be summarized as:

Student / Administrator





This architecture ensures that users do not directly access the database. All major requests pass through the backend where validation, authorization, and business logic can be applied.

4.17 Summary

The system architecture provides a structured and modular design for the Smart Library Database System and Student Services Portal.

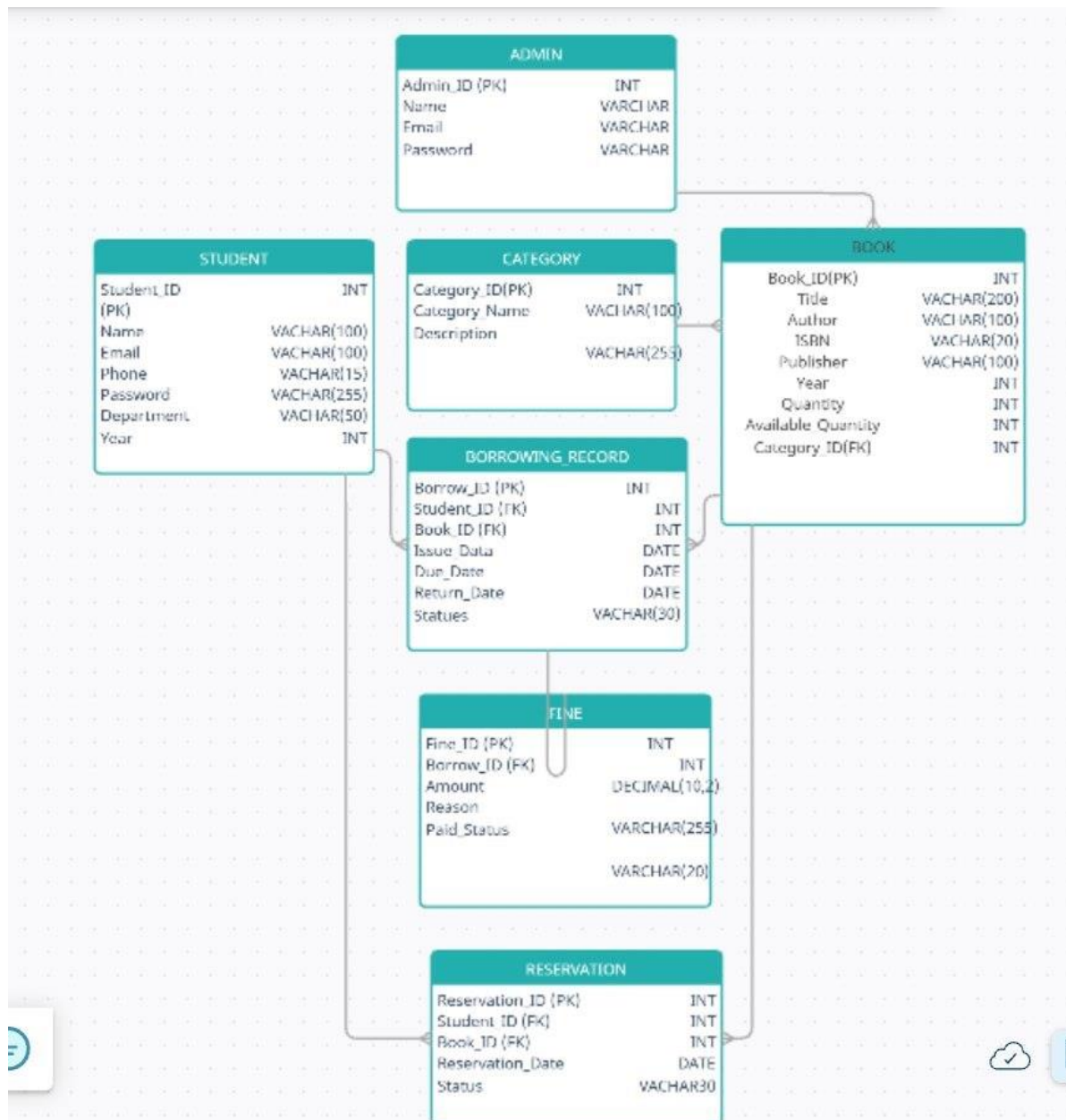
The React.js frontend provides the user interface, while Node.js and Express.js manage application logic and REST APIs. MySQL provides persistent and structured data storage.

Authentication and authorization help protect user functions, while the Member Management, Book and Asset Management, Borrowing, Reservation, Fine, Search and Reports, Member History, and Admin Management modules provide the major functionality of the system.

Offline caching, background synchronization, and notifications improve the accessibility and usability of the platform.

Overall, the architecture provides a foundation for developing a maintainable, organized, and extensible digital library system.

CHAPTER -5 DESIGN



Section 6.1 – Module-Wise Implementation

The implementation of the **Smart Library Database System and Student Services Portal** was structured into a set of interconnected modules that collectively provide an efficient digital platform for library management and student services. Each module was designed with a specific functional responsibility while maintaining integration with the overall system through a centralized database and backend API architecture. This modular approach improves scalability, maintainability, data consistency, and ease of use. The system provides students with convenient access to library services such as registration, login, book searching, availability checking, borrowing, returning, and tracking borrowing history. At the administrative level, librarians can manage books, students, categories, and borrowing records through a dedicated administration interface. The implementation focuses on reducing manual record keeping, improving data accuracy, and providing faster and more accessible library services.

The **Student Registration and Authentication Module** is one of the fundamental components of the system. It allows students to create an account and securely log in to access library services. During registration, the required student information is collected and stored in the MySQL database. The login functionality verifies the user's credentials before providing access to the student portal. Authentication ensures that only registered users can access personalized services and borrowing information. The module is integrated with the backend through API endpoints, allowing authentication-related requests to be processed securely. This module provides the foundation for personalized student services and ensures that library transactions can be associated with the correct student account.

The **Book Search and Filtering Module** provides students with an efficient way to find books available in the library. Users can search for books based on relevant information such as title, author, category, or other available book details. Filtering functionality helps students narrow down the search results and locate required resources more quickly. Each search result can provide important information about the selected book, including its details and availability status. The module communicates with the backend to retrieve the required book information from the MySQL database. By replacing manual searching through physical library records with a digital search mechanism, the system improves the speed and convenience of accessing library resources.

The **Book Details and Availability Module** provides detailed information about individual books stored in the database. When a student selects a book, the system retrieves its corresponding information through the backend API and displays it through the frontend interface. Availability information allows students to determine whether a particular book can currently be borrowed. The module works closely with the borrowing and return functionality so that changes in book availability are reflected in the system. This integration helps maintain accurate records and prevents confusion regarding the current status of library resources.

The **Borrow and Return Management Module** manages the main library transaction process. Students can borrow available books and return them through the student portal. When a borrowing transaction takes place, the system records the relevant student, book, and transaction information in the database. The availability status of the book is also updated accordingly. When a book is returned, the corresponding borrowing record is updated and the book can again become available for other students. This module reduces the need for manual transaction recording and improves the accuracy of library borrowing information. It also provides a structured connection between students, books, and borrowing records within the database.

The **Borrowing History Module** allows students to view their previous and current borrowing records. The system retrieves transaction information associated with the logged-in student and presents it through the portal. This allows students to track the books they have borrowed and review their library activity. The information is obtained from the centralized MySQL database through backend API requests. Maintaining borrowing history also helps the system preserve a clear record of library transactions and supports better management of student activities.

The **Admin and Library Management Module** provides the administrative functionality required for maintaining the library database. Librarians or administrators can perform CRUD operations such as **adding, updating, viewing, and deleting books**. They can also manage student information, book categories, and borrowing records. The admin interface provides centralized control over the major entities of the system. Changes made by administrators are processed through the Node.js and Express.js backend and stored in the MySQL database. This module reduces manual database maintenance and provides an organized method for managing library resources.

The **Student Management Module** allows administrators to manage student information stored in the system. Administrators can view registered students and maintain their associated records. Since student information is connected with borrowing transactions, proper management of student records is important for maintaining database consistency. The module uses backend APIs to perform the required operations and communicates with the MySQL database for persistent storage.

The **Book Category Management Module** organizes books into meaningful categories. Administrators can create, update, and manage categories according to the library's requirements. Categorization improves the organization of the book collection and supports the filtering and searching functionality available to students. The relationship between categories and books is maintained within the database, allowing the frontend to display organized and relevant search results.

The **Database and Transaction Management Module** acts as the central data layer of the application. MySQL is used to store and manage information related to users, students, books, categories, and borrowing transactions. The database provides structured storage and maintains relationships between

different entities. The backend communicates with the database to perform operations such as inserting new records, retrieving information, updating existing records, and deleting records where required. This centralized approach improves data consistency and makes the system easier to maintain compared with separate manual records.

The **Backend API Module** acts as the communication layer between the React.js frontend and MySQL database. Node.js with Express.js is used to implement the server-side logic and API endpoints.

Whenever a user performs an operation through the frontend, such as searching for a book or borrowing a book, the corresponding request is sent to the backend. The backend processes the request, performs the required database operation, and sends the appropriate response back to the frontend. This separation between presentation, business logic, and data storage improves the organization and maintainability of the system.

The **Student Services Portal** integrates the major student-oriented functionalities into a single web interface. Instead of requiring students to depend completely on physical library processes, the portal allows them to access important library services digitally. Registration, login, book search, book availability, borrowing, returning, and borrowing history are brought together within the same system. The unified interface makes library services more accessible and provides a convenient user experience.

Collectively, the modular implementation of the **Smart Library Database System and Student Services Portal** creates a unified digital library environment. Each module performs a specific responsibility while communicating with other modules through APIs and shared database structures. The modular architecture makes it possible to maintain or improve individual components without significantly affecting the complete system. Future extensions such as additional student services, advanced search functionality, notifications, or expanded reporting can also be incorporated without completely redesigning the existing architecture. Overall, the implementation demonstrates the practical integration of **full-stack web development, database management, authentication, CRUD operations, API communication, and system integration** in a real-world library management environment.

Section 6.2 – Front-End Logic (UI Rendering and State Management)

The front-end of the **Smart Library Database System and Student Services Portal** provides the primary interaction layer between students, administrators, and the system. Since the application combines library management with student-oriented services, the front-end was designed to present complex database operations through a simple and accessible web interface. The main objective of the frontend is to allow students to perform activities such as registration, login, book searching, viewing book details, checking availability, borrowing and returning books, and viewing borrowing history without requiring direct interaction with the underlying database. Administrators are provided with interfaces for managing books, students, categories, and borrowing records. React.js is used to implement the frontend and create a structured, interactive user experience.

The foundation of the frontend is based on a **component-based architecture** provided by React.js. The interface is divided into reusable components responsible for individual functions such as navigation, authentication forms, book search, book listings, book details, borrowing information, and administrative management. This approach avoids unnecessary duplication of interface logic and makes the application easier to maintain. Individual components can be modified or extended independently while continuing to work with the rest of the application. The component-based structure also allows common interface elements to be reused across multiple pages of the student and administrator portals.

The **UI rendering logic** determines how information retrieved from the backend is displayed to users. When a student searches for a book, the frontend sends the search request to the appropriate backend API. After receiving the response, React renders the returned book information in the relevant section of the interface. Similarly, when a student opens a book's details, the frontend dynamically displays information such as the book details and availability status. This process allows the interface to update according to user actions rather than requiring users to interact directly with database records.

The **state management mechanism** is responsible for maintaining the current state of the application and synchronizing user actions with the displayed information. Important application states can include logged-in user information, search input, book search results, selected book details, availability information, borrowing records, and administrative data. When a user performs an action, the corresponding state is updated and the React interface re-renders the affected component. This allows the system to provide updated information without unnecessarily reloading the complete application.

Authentication state is particularly important in the student services portal. After a successful login, the frontend maintains the relevant user session information so that personalized services can be displayed. The interface can distinguish between student-oriented functionality and administrative functionality based on the authenticated user's role. This helps ensure that library management operations are presented only within the appropriate administrative interface while students receive access to their own library services.

The **book search interface** uses frontend state to capture the user's search and filtering requirements. When the user enters search information or selects a filtering option, the frontend processes the input and communicates with the backend API to retrieve the relevant records. The returned data is then rendered dynamically as book results. This creates a responsive search experience and reduces the effort required to locate resources within the library database.

The **borrowing and returning interface** also depends on coordinated state updates. When a student borrows a book, the frontend sends the borrowing request to the backend. After the backend successfully updates the database, the frontend receives the response and updates the displayed book availability and borrowing information. The same process occurs when a book is returned. This synchronization ensures that the information displayed to students reflects the latest available database information.

The **admin interface** uses reusable React components to support CRUD operations. Administrators can access forms for adding or updating books and interfaces for viewing and managing students, categories, and borrowing records. When an administrator submits a change, the frontend sends the required data to the corresponding Express.js API endpoint. After the backend performs the database operation, the frontend updates the displayed information accordingly. This creates a clear flow between user interaction, API communication, database processing, and UI rendering.

The frontend communicates with the backend using **API-based interaction**. React.js acts as the presentation layer, while Node.js and Express.js handle server-side processing. The frontend does not directly manipulate the MySQL database. Instead, requests are sent to backend API endpoints, which process the request and communicate with MySQL. The resulting data is then returned to the frontend and displayed to the user. This architecture provides a clear separation between the user interface, application logic, and database layer.

The navigation structure of the portal is designed around the main activities performed by different users. Students can move between services such as the dashboard, book search, book details, borrowing and return functionality, and borrowing history. Administrators can navigate between book management, student management, category management, and borrowing record management. Organizing the navigation according to user roles and functions makes the system easier to understand and reduces unnecessary complexity.

The frontend also provides feedback based on the result of user operations. For example, after a successful login, borrowing operation, return operation, or administrative update, the interface can reflect the updated system state. Similarly, when an operation cannot be completed, the interface can display an appropriate response to the user. Such feedback improves usability and helps users understand whether their requested operation was successfully processed.

Overall, the front-end logic of the **Smart Library Database System and Student Services Portal** provides a structured connection between users and the underlying database system. The combination of **React.js component-based development, dynamic UI rendering, state management, authentication, API communication, and role-based interfaces** creates an integrated and user-friendly platform. The frontend works together with the Node.js and Express.js backend and MySQL database to provide students with convenient library services while giving administrators efficient tools for managing library resources and records.

6.3 API Integration

The **API Integration** stage forms the communication backbone of the **Smart Library Database System and Student Services Portal**, connecting the React.js frontend with the Node.js and Express.js backend and the MySQL database. Application Programming Interfaces (APIs) allow the frontend to send requests to the backend and receive the required data without directly accessing the database. This approach provides a clear separation between the presentation layer, application logic, and data layer while allowing different modules of the system to communicate efficiently.

The system primarily uses **internal RESTful APIs** to manage library and student-related operations. These APIs are implemented using **Node.js and Express.js** and are organized according to the major entities and functionalities of the application. The main API operations include user authentication, book management, student management, category management, and borrowing and returning of books. Each API endpoint performs a specific operation and communicates with the MySQL database through the backend.

For example, authentication-related endpoints handle student registration and login. Book-related endpoints are responsible for retrieving book information, adding new books, updating existing book details, and deleting books when required. Student-related endpoints allow administrators to retrieve and manage registered student information. Category-related endpoints are used to organize books into different categories. Borrowing-related endpoints manage book issue and return operations and maintain the corresponding borrowing records.

The communication between the React.js frontend and Node.js backend is performed through **HTTP requests and JSON data**. When a student searches for a book, the frontend sends a request to the appropriate backend API. The Express.js server receives the request, processes the required logic, and retrieves the relevant information from MySQL. The server then returns the result in JSON format, which is received by the React frontend and displayed dynamically to the student. This process allows the interface to update according to the latest database information without requiring direct database access from the client side.

The API architecture follows a structured request-response model. Common HTTP methods are used according to the type of operation being performed. **GET** requests are used to retrieve information such as book details, student records, categories, and borrowing history. **POST** requests are used for operations such as student registration and creating borrowing records. **PUT/PATCH** requests can be used to update existing book or student information, while **DELETE** requests are used for removing records where permitted. This separation makes the API structure easier to understand and maintain.

The **Book Management API** is one of the major API groups in the system. It provides communication between the admin interface and the MySQL database for managing library books. When an administrator adds a new book, the frontend sends the book information to the backend. The backend validates the request and inserts the corresponding record into the database. Similarly, update and delete requests are processed by the backend and the resulting changes are reflected in the frontend.

The **Book Search and Availability API** supports the student portal. When a student enters a search term or applies a filter, the frontend sends the relevant request to the backend. The backend searches the appropriate database records and returns matching books. The response contains the required book information and availability status, allowing students to quickly determine whether a book can be borrowed.

The **Borrowing and Return API** handles the transaction between students and books. When a student borrows an available book, the frontend sends the required student and book information to the backend. The backend creates the borrowing record in MySQL and updates the availability status of the book.

During the return process, the backend updates the borrowing record and changes the book's availability accordingly. This ensures that the borrowing information and book status remain synchronized.

The **Student Management API** allows the administrator to interact with student information stored in the database. Through the admin portal, administrators can retrieve and manage registered student records. The API acts as an intermediary between the React interface and the MySQL database, ensuring that database operations are performed through the server rather than directly from the client.

The **Category Management API** is used to manage the categories associated with books. Administrators can create, update, retrieve, and manage categories through the admin interface. The frontend sends the relevant request to the Express.js backend, which performs the corresponding database operation. Categories can then be used by the book search and filtering functionality to help students find resources more efficiently.

Authentication and authorization are important parts of API integration. The system verifies user credentials during login and ensures that protected operations are available only to authorized users. Student-specific operations, such as viewing personal borrowing history, are associated with the authenticated student account. Administrative operations such as adding, updating, or deleting books and managing borrowing records are restricted to the appropriate administrator or librarian role. This provides controlled access to sensitive library management functions.

The API layer also performs **input validation** before sending information to the database. User-provided information such as registration details, book information, and borrowing requests is checked by the backend before the corresponding database operation is performed. Validation helps prevent invalid records and improves the reliability and consistency of the stored data.

Error handling is implemented at the API level to ensure that unexpected situations are handled properly. If a requested book does not exist, a borrowing operation is invalid, login credentials are incorrect, or a database operation fails, the backend returns an appropriate response to the frontend. HTTP status codes such as **200 (OK)**, **201 (Created)**, **400 (Bad Request)**, **401 (Unauthorized)**, **404 (Not Found)**, and **500 (Internal Server Error)** can be used to indicate the result of an API request. The frontend can then display suitable feedback to the user.

Database consistency is particularly important for borrowing and returning operations. Since a borrowing transaction can involve both the creation or modification of a borrowing record and the update of book availability, the backend must process these related operations carefully. This ensures that the system does not incorrectly show a book as available when it has already been borrowed. The API layer therefore acts as an important control point between user actions and database transactions.

The API integration also supports **CRUD operations**, which are central to the administration functionality of the project. CRUD represents **Create, Read, Update, and Delete** operations. These operations are applied to major entities such as books, students, categories, and borrowing records. By implementing CRUD functionality through APIs, administrators can manage the library database through the web interface without directly modifying database tables.

The backend API architecture provides a clear separation of responsibilities. **React.js** handles the user interface and user interactions, **Node.js** and **Express.js** handle server-side logic and API requests, and **MySQL** provides persistent storage for library and student information. This three-layer interaction improves maintainability and allows each layer to be developed or modified independently.

API testing can be performed using tools such as **Postman** to verify that individual endpoints return the expected results. Testing can include valid requests, invalid input, missing information, authentication failures, and database-related errors. The responses are checked to ensure that the API returns appropriate data and status codes. This helps identify communication and backend logic problems before the complete application is used by students or administrators.

Overall, the API integration strategy provides the communication layer that connects all major components of the **Smart Library Database System and Student Services Portal**. It enables smooth data exchange between the React.js frontend, Node.js/Express.js backend, and MySQL database. Through RESTful APIs, authentication, CRUD operations, validation, error handling, and structured JSON communication, the system provides a reliable and organized method for managing library resources and student services. The modular API architecture also makes it possible to extend the system in the future with additional features such as **notifications, advanced search, fine management, reservation functionality, reports, and additional student services** without requiring a complete redesign of the existing system.

6.4 Authentication and Authorisation

The **Authentication and Authorisation** module is an important security component of the **Smart Library Database System and Student Services Portal**. It is responsible for verifying the identity of users, protecting personal and library-related information, and controlling access to different functionalities of the system. Authentication determines whether a user is a registered student or administrator, while authorisation determines which operations that user is allowed to perform. Since the system handles student details, book information, and borrowing records, controlled access is essential for maintaining data security and integrity.

The system provides a **registration and login mechanism** through which students can create accounts and access the Student Services Portal. During registration, required student information and login credentials are collected through the React.js frontend and sent to the Node.js and Express.js backend through API requests. The backend validates the received information and stores the appropriate user record in the MySQL database. During login, the credentials provided by the student are verified against the stored account information. Only users with valid credentials are allowed to access the protected student services.

Authentication is integrated with the overall API architecture of the application. When a user submits the login form, the React frontend sends the credentials to the appropriate authentication API endpoint. The Express.js backend processes the request and communicates with the MySQL database to verify the user's account. If the credentials are valid, the backend returns a successful authentication response to the frontend. The frontend can then provide access to the student's dashboard and other authorized library services.

The system also implements **role-based authorisation** to distinguish between students and administrators. Students are provided access to services required for normal library usage, such as searching and filtering books, viewing book details and availability, borrowing books, returning books, and viewing their borrowing history. Administrators or librarians have additional privileges because they are responsible for managing the library database. Their functions include adding, updating, and deleting books, managing students, managing book categories, and managing borrowing records.

The distinction between student and administrator permissions helps prevent unauthorized users from performing administrative operations. For example, a student should not be able to access the interface for deleting books or modifying student records. Such operations are restricted to authorized administrators. This ensures that important library information remains under controlled management.

At the **frontend level**, React.js can use conditional rendering and protected navigation to display appropriate features according to the user's role. When a student logs in, the interface displays student-oriented services, while an administrator receives access to the administration dashboard. Administrative options such as **Add Book, Update Book, Delete Book, Manage Students, Manage Categories, and Manage Borrowing Records** are therefore separated from regular student functionality.

However, frontend restrictions alone are not sufficient for security. The **backend also verifies authorization** before processing protected API requests. When an administrative operation is requested, the Express.js server checks whether the authenticated user has the required privileges. If a student attempts to access an administrative endpoint, the backend rejects the request even if the user attempts to access that endpoint directly. This provides an additional security layer between users and the database.

The authentication mechanism is closely connected with the **MySQL database**. User and student account information is maintained in the database, while borrowing records are associated with the relevant student accounts. This relationship allows the system to identify which student performed a particular library transaction. As a result, students can view their own borrowing history while administrators can access the records required for library management.

Password security is also an important consideration within the authentication process. User passwords should not be stored in plain text. Instead, passwords can be securely hashed before being stored in the MySQL database. During login, the password provided by the user is compared with the stored hashed value through the backend authentication process. This reduces the risk associated with exposing actual passwords in the database.

The system also performs **input validation** during registration and login. Required fields are checked before the information is processed by the backend. Invalid or incomplete information is rejected and an appropriate response is returned to the frontend. Validation helps prevent incorrect account information from being stored and improves the reliability of the authentication process.

Session or token-based authentication can be used to maintain the authenticated state of the user while interacting with protected services. After successful login, the authentication information is maintained by the application and used when accessing protected API operations. This allows the backend to determine whether the request originates from an authenticated user and whether the requested operation is permitted.

Error handling is incorporated into the authentication module to provide appropriate responses for different situations. If a user enters incorrect login credentials, the system can return an authentication error. If a user attempts to access a resource without the required authorization, the backend can return an appropriate authorization response. Common HTTP status codes such as **401 (Unauthorized), 403**

(Forbidden), 400 (Bad Request), and 500 (Internal Server Error) can be used to communicate authentication and server-related errors.

The authentication system also protects the **borrowing and returning functionality**. A student must be authenticated before performing library transactions associated with their account. When a borrowing request is submitted, the backend can associate the transaction with the authenticated student's information. This prevents anonymous users from creating unauthorized borrowing records and ensures that transactions are linked to the correct student.

Similarly, access to the **admin management functions** is protected through authorization controls. Only authorized administrators can modify important database information such as books, students, categories, and borrowing records. This helps prevent accidental or unauthorized changes that could affect the accuracy of the library database.

Security is also maintained during communication between the frontend and backend. In a deployed environment, **HTTPS** can be used to encrypt communication between the React.js frontend and Node.js/Express.js backend. This protects sensitive information such as login credentials and user data while it is being transmitted between the client and server.

The authentication and authorization mechanism also supports the principle of **least privilege**. Users receive only the permissions required for their role. Students receive access to services required for borrowing and managing their own library activities, while administrators receive additional permissions necessary for managing library resources. This separation reduces the possibility of unauthorized database operations.

From the administrative perspective, the authorization system provides controlled access to the library management dashboard. Administrators can manage books, students, categories, and borrowing records through the appropriate interfaces. Since these operations can directly affect the database, restricting them to authorized personnel helps maintain the integrity and consistency of the system.

Overall, the **Authentication and Authorisation module** provides an important security layer for the **Smart Library Database System and Student Services Portal**. By combining user registration, secure login, role-based access control, protected APIs, password protection, input validation, and controlled administrative access, the system ensures that library and student information is accessed only by appropriate users. The module works together with **React.js, Node.js, Express.js, and MySQL** to provide a secure and organized environment for digital library services while maintaining the confidentiality, integrity, and reliability of the stored information.

6.5 Authentication Implementation

The **Authentication Implementation** is an important part of the Smart Library Database System and Student Services Portal because it controls access to the application and protects student and library-related information. The authentication system is designed to ensure that only registered and valid users can access the services provided by the platform. The implementation begins with the **student registration process**, where a new student provides the required information through the registration interface. The React.js frontend collects the entered information and sends it to the Node.js and Express.js backend through an API request. The backend receives the request, validates the provided information, checks whether the student account already exists, and then stores the required account details in the MySQL database. This process creates a unique account for each student and allows the student to use the system without depending on manual library registration procedures.

The **login functionality** is implemented as the main authentication mechanism for accessing the Student Services Portal. When a registered student enters their login credentials, the React.js frontend sends the information to the authentication API implemented using Node.js and Express.js. The backend receives the credentials and verifies them against the corresponding user information stored in the MySQL database. If the provided credentials are valid, the authentication process is considered successful and the student is granted access to the appropriate portal. If the credentials are incorrect, the backend returns an error response and the frontend displays an appropriate message to the user. This process prevents unauthorized users from directly accessing student-specific library services and ensures that personal information such as borrowing history is associated with the correct account.

The authentication implementation is closely connected with the **role-based access structure** of the system. The application primarily supports two types of users: **students and administrators/librarians**. Students use the system to access library services, whereas administrators have additional permissions required for managing the library database. After authentication, the system determines the user's role and provides access to the corresponding interface. A student can access functionalities such as searching for books, filtering books, viewing book details, checking availability, borrowing books, returning books, and viewing borrowing history. Administrators, on the other hand, can access management functionality such as adding books, updating book information, deleting books, managing students, managing categories, and managing borrowing records. This separation ensures that regular students cannot perform operations that could modify important library information.

The authentication process also provides **protected access to API endpoints**. The frontend does not directly communicate with the MySQL database. Instead, all important operations are sent through the

Node.js and Express.js backend. Before processing protected requests, the backend can verify whether the user is authenticated and whether the user has the required role or permission. For example, when a student attempts to borrow a book, the backend verifies the authenticated student's identity before creating the borrowing record. Similarly, when an administrator attempts to delete or update a book, the backend verifies that the request is coming from an authorized administrator. This provides security at the server level and prevents users from bypassing frontend restrictions by directly sending requests to backend APIs.

The implementation also considers **password security**. Passwords should not be stored directly in plain-text form in the database. Instead, the authentication layer can use secure password hashing techniques before storing credentials in MySQL. During login, the password entered by the user is processed and compared with the stored secure value. This approach helps protect user credentials even if the database is accessed without authorization. Input validation is also performed during registration and login to ensure that required fields are provided and that invalid information is not unnecessarily processed or stored.

The authentication system is integrated with the **React.js state management and user interface**. Once the user successfully logs in, the frontend maintains the authenticated state and displays the appropriate dashboard. Components such as the navigation bar, student dashboard, book search page, borrowing section, and admin management pages can respond according to the user's authentication status. If the user is not authenticated, protected pages can redirect the user to the login interface rather than displaying private information. When the user logs out, the authenticated state is cleared and the user is returned to the login or public section of the application. This provides a clear separation between authenticated and unauthenticated areas of the system.

The authentication implementation is particularly important for the **borrowing history functionality**. Since borrowing records contain information about which books were borrowed by a particular student, this information should only be displayed to the appropriate authenticated student or authorized administrator. When a student opens the borrowing history section, the system identifies the logged-in student and retrieves the corresponding borrowing records from the backend. The backend communicates with the MySQL database and returns only the relevant records. This prevents students from viewing another student's borrowing information and maintains privacy within the library system.

Authentication is also integrated with the **book borrowing and returning workflow**. When a student selects an available book and performs a borrowing operation, the request is associated with the student's authenticated account. The backend processes the request and creates the appropriate borrowing record in the database. When the student returns the book, the system identifies the associated transaction and updates the record. This ensures that every major library transaction has a connection with an authenticated user account. As a result, the database maintains a more reliable history of library activity.

Error handling forms another important part of the authentication implementation. The system handles situations such as incorrect credentials, duplicate registration, missing input fields, unauthorized access, invalid requests, and server-side failures. Appropriate responses are sent from the Express.js backend to the React frontend, allowing the interface to provide understandable feedback to the user. For example, an incorrect login attempt can result in a message asking the user to verify their credentials, while an unauthorized attempt to access an administrative operation can be rejected by the backend. This improves both system security and user experience.

The authentication implementation therefore acts as a security gateway for the complete **Smart Library Database System and Student Services Portal**. It connects the React.js frontend, Express.js backend, and MySQL database while ensuring that users are properly identified before accessing protected services. Through registration, login, password protection, role-based authorization, protected API endpoints, authentication state management, and controlled access to borrowing information, the system provides a secure environment for students and administrators. The implementation also creates a foundation for future security improvements such as email verification, password reset functionality, multi-factor authentication, account activity tracking, and more detailed administrator permissions.

6.6. Notification Implementation

The **Notification Implementation** is designed to improve communication between the Smart Library Database System and its users by providing timely information about important library activities. Notifications can be used to inform students about events associated with their accounts, books, and borrowing transactions, while administrators can receive information related to changes in library records and system activities. Instead of requiring users to continuously check different sections of the portal for updates, the notification functionality provides important information directly through the application interface. The notification mechanism is integrated with the existing React.js frontend, Node.js and Express.js backend, and MySQL database so that notifications can be generated based on relevant system events.

The main purpose of the notification system is to provide **real-time or event-based feedback** after important operations are performed. For example, when a student successfully logs in, borrows a book, returns a book, or performs another supported library operation, the system can generate a corresponding notification. Similarly, when an administrator successfully adds, updates, or deletes a book, the system can provide confirmation that the requested operation has been completed. These notifications help users understand the result of their actions and reduce uncertainty while interacting with the portal. The notification information can be displayed through notification messages, alerts, or a dedicated notification section within the student or administrator dashboard.

The notification mechanism is connected to the **backend API layer** of the application. When an important event occurs, the Node.js and Express.js backend processes the event and can generate an appropriate notification. For example, after a successful borrowing transaction, the backend can create a notification containing information about the transaction and associate it with the corresponding student account. The notification can then be stored in the MySQL database so that it remains available to the user even after navigating to another page. The React.js frontend retrieves the notification information through an API and displays it within the appropriate section of the portal.

One of the major uses of the notification system is related to **book borrowing and returning**. When a student successfully borrows a book, the system can generate a notification confirming the borrowing operation. The notification can contain relevant information such as the book title, transaction status, and borrowing-related details. Similarly, after a book is successfully returned, the system can notify the student that the return operation has been completed. These notifications provide confirmation and help students keep track of their library activities without manually checking the borrowing records after every transaction.

Notifications can also be used for **book availability-related information**. When students search for books, the system displays their availability through the book management and search modules. The notification functionality can complement this process by providing updates when relevant book status changes occur. For example, a system extension can notify a student when a previously unavailable book

becomes available. This can improve the usefulness of the Student Services Portal by making library information more accessible and reducing the need for repeated manual checking.

The notification system can also provide **administrative notifications**. Administrators are responsible for maintaining books, students, categories, and borrowing records within the library database. The system can generate notifications when important administrative operations are completed successfully. For example, after a new book is added, the interface can display a confirmation notification. Similar feedback can be generated after updating or deleting a book, modifying a category, or completing an administrative database operation. These notifications provide immediate feedback and help administrators confirm that their actions have been processed successfully.

The **React.js frontend** is responsible for presenting notification information to users in an understandable manner. A notification component can be integrated into the dashboard or navigation area of the application. When new notifications are available, the interface can display a notification indicator or message. Selecting the notification section can allow the user to view the available notifications and understand the event associated with each message. The frontend can also update the notification display when new information is received from the backend. This creates a more interactive experience and ensures that important system information is visible to users.

The notification data can be maintained in the **MySQL database** as part of the overall application data architecture. A notification record can be associated with a particular user and can contain information such as the notification message, notification type, creation time, and read/unread status. Associating notifications with user accounts ensures that each student receives information relevant to their own activities. It also allows the system to maintain a history of notifications rather than displaying information only temporarily. Administrators can similarly receive notifications relevant to administrative activities if such functionality is enabled.

The **read and unread notification mechanism** helps users identify whether important information has already been viewed. When a new notification is generated, it can initially be marked as unread. Once the user opens or acknowledges the notification, its status can be updated through the backend API. The corresponding record in MySQL can then be changed from unread to read. This provides a simple but effective mechanism for tracking user interaction with system messages. A notification counter or indicator can also be displayed on the dashboard to inform users when new notifications are available.

The notification module is integrated with the **API communication architecture** of the system. The React frontend sends requests to the Express.js backend to retrieve notifications, mark notifications as read, or perform other supported notification operations. The backend processes these requests and interacts with MySQL to retrieve or update notification records. The resulting information is returned to the frontend in JSON format. This follows the same architecture used by other modules of the project and

maintains a consistent separation between the user interface, business logic, and database layers.

Notification implementation also improves **user experience and system transparency**. For example, if a borrowing operation fails because a book is no longer available, the system can provide a notification or alert explaining the situation rather than simply failing silently. Similarly, if an administrator successfully updates a book record, a confirmation notification provides immediate feedback. These responses help users understand what is happening within the system and make the application feel more responsive and reliable.

The notification functionality can also support **future library services**. As the system is extended, notifications can be used for reminders related to borrowing periods, upcoming return dates, overdue books, reservation availability, library announcements, or other student services. Such functionality can be connected to the existing borrowing records and user information stored in MySQL. The modular API-based design allows these features to be introduced without changing the fundamental architecture of the application.

Overall, the **Notification Implementation** provides an additional communication layer within the Smart Library Database System and Student Services Portal. It connects important system events with clear user feedback and allows students and administrators to remain informed about relevant activities. By integrating notification generation with the **React.js frontend, Node.js and Express.js backend, and MySQL database**, the system can provide structured event-based communication for borrowing, returning, book management, authentication-related events, and other library services. The notification module therefore contributes to a more interactive, transparent, and user-friendly digital library environment while also providing a foundation for future features such as automated reminders, overdue alerts, reservations, and extended student services.

CHAPTER 7- FEATURES

7.1 List of Main Features

The Smart Library Database System and Student Services Portal has been designed as a single, unified digital environment that brings together every core library operation and student-facing service into one accessible platform. Rather than treating registration, searching, borrowing, and administration as separate, disconnected tasks, the system links them through a common React.js interface on the front end and a shared Node.js and Express.js API layer on the back end, all backed by a single MySQL database. Each feature described below therefore functions both as a standalone capability for the end user and as a coordinated part of a larger workflow that spans the student portal and the administrative dashboard.

Student Registration and Login. New students can create an account by providing their basic academic and contact details, after which they can log in securely to access the portal. This is the entry point to every other service in the system and establishes the identity used to track borrowing activity.

Book Search and Filter. Students can search the library catalogue by title, author, ISBN, or category, and can further refine results using filters such as availability, subject, or publication year. This reduces the time spent browsing physical shelves or unstructured lists.

Book Details and Availability. Selecting a book opens a detailed view showing its description, author, category, number of copies, and real-time availability status, so students always know whether a title can be borrowed before making a trip to the library.

Borrow and Return. Students can request to borrow an available book and later mark it as returned, with the system automatically updating the number of available copies and recording due dates to help prevent overdue items.

Borrowing History. A personal borrowing history page lists every book a student has previously borrowed or currently holds, along with issue and return dates, giving students a simple way to track their own reading and borrowing patterns.

Student Dashboard and Profile. After logging in, students land on a dashboard summarising their currently borrowed books, due dates, and quick links to search and history, with an accompanying profile section for updating personal details.

Admin Book Management. Librarians can add new books to the catalogue, update existing details such as stock count or category, and remove books that are no longer in circulation, all through a dedicated administrative interface.

Admin Student and Category Management. The admin portal allows librarians to manage student records and

maintain the list of book categories, keeping the catalogue organised and the student database accurate and up to date.

Admin Borrowing Records Management. Librarians can view, search, and manage all borrowing transactions across the library, including overdue items, enabling proactive follow-up and accurate reporting on circulation activity.

Authentication and Role-Based Access. The system distinguishes between student and administrator roles, ensuring that sensitive management functions are only available to authorised library staff while students retain access to their own services.

API-Driven Frontend-Backend Communication. Every action performed in the React.js interface is carried out through calls to the Express.js API, which validates and processes the request before reading from or writing to the MySQL database, keeping the presentation layer and the data layer cleanly separated.

Collectively, these features fulfil the project's central objective: to replace manual, paper-based library record keeping with a fast, accurate, and accessible digital system. Each capability, though independently useful, is connected through shared authentication, a shared database schema, and a consistent API layer, so that the student experience of searching and borrowing and the administrative experience of managing the catalogue remain in constant, reliable synchronisation.

7.2 How Each Feature Works

The following section describes, feature by feature, how the Smart Library Database System behaves from the perspective of the person using it and how that behaviour is implemented underneath, across the React.js front end, the Node.js and Express.js back end, and the MySQL database.

1. Student Registration and Login Feature

User Perspective:

A new student opens the registration page and enters details such as name, roll number, email, and a password. On submitting the form, the system either creates the account and redirects the student to the login page, or displays a clear validation message if any required field is missing or already in use. Returning students log in with their email and password, after which they are taken directly to their personal dashboard.

Technical Description:

The registration form sends a POST request to an Express.js endpoint, which validates the input, hashes the password using bcrypt, and inserts a new record into the students table in MySQL. The login endpoint verifies the submitted credentials against the stored hash and, on success, issues a JSON Web Token (JWT) that the React.js front end stores and attaches to subsequent API requests to identify the logged-in student.

2. Book Search and Filter Feature

User Perspective:

From the search page, a student types a keyword into the search bar or selects a category from a dropdown filter. Results update to show matching books with their cover information, author, and availability, allowing the student to narrow down a large catalogue to a short, relevant list within seconds.

Technical Description:

The search bar triggers a GET request to a books endpoint with query parameters for the search term and any selected filters. The Express.js route builds a parameterised SQL query against the books table in MySQL, using LIKE conditions for text search and WHERE clauses for category or availability filters, and returns the matching rows as JSON for the React.js interface to render.

3. Book Details and Availability Feature

User Perspective:

Clicking on any book in the search results opens a details page showing the full description, author, category, total copies, and how many copies are currently available to borrow. If no copies remain, the interface clearly marks the book as unavailable rather than allowing a borrow request to proceed.

Technical Description:

The details view calls a books-by-id endpoint that retrieves a single row from the books table, including a computed available_copies value derived from comparing total_copies against the count of currently unreturned borrowing records. The React component conditionally renders a disabled borrow button when available_copies reaches zero.

4. Borrow and Return Feature

User Perspective:

When a student chooses to borrow an available book, a confirmation dialog shows the due date before the request is submitted. Once borrowed, the same book appears in the student's dashboard with a return button; clicking it marks the book as returned and immediately updates its availability for other students.

Technical Description:

Borrowing creates a new row in a borrow_records table linking student_id and book_id, along with issue and due dates, inside a database transaction that also decrements the book's available copy count. Returning a book updates that record's return_date and increments the copy count back, ensuring the two operations never leave the data in an inconsistent state even if a request fails partway through.

5. Borrowing History Feature

User Perspective:

The history page lists every book a student has ever borrowed, sorted from most to least recent, showing the issue date, due date, and return date for each entry, along with a status label such as Returned, Active, or Overdue.

Technical Description:

This page calls an endpoint that joins the borrow_records table with the books table on book_id, filtered by the logged-in student's ID from the JWT. The Express.js layer computes the status of each record by comparing the due date and return date against the current server date before sending the results as JSON.

6. Student Dashboard and Profile Feature

User Perspective:

After logging in, the student is shown a dashboard summarising currently borrowed books and any items approaching their due date, alongside a profile section where the student can view and update contact details such as email or phone number.

Technical Description:

The dashboard combines data from two endpoints: an active-borrows query filtered to records with no return_date, and a student-profile endpoint that reads from the students table. Profile updates are submitted as a PUT request that validates the new values before writing them back to the corresponding row in MySQL.

7. Admin Book Management Feature

User Perspective:

Librarians log in to a separate admin view where they can add a new book by filling in its title, author, category, and copy count, edit any of these fields for an existing book, or remove a book entirely from the catalogue.

Technical Description:

Add, update, and delete actions map to POST, PUT, and DELETE routes on the books resource, each protected by middleware that checks the admin role encoded in the JWT before allowing the operation to reach the database. All three operations act directly on the books table, with foreign-key constraints preventing deletion of a book that still has active, unreturned borrowing records.

8. Admin Student and Category Management Feature

User Perspective:

From the admin portal, librarians can view the full list of registered students, search for a specific student, and manage the set of categories used to classify books, adding new categories as the collection grows or retiring ones that are no longer needed.

Technical Description:

Student records are exposed through admin-only GET and PUT routes on the students resource, while categories

are managed through a small categories table referenced by a foreign key on the books table, so that renaming or removing a category is reflected consistently across the catalogue.

9. Admin Borrowing Records Management Feature

User Perspective:

Librarians can open a records screen that lists every borrowing transaction in the system, filterable by student, book, or status, making it straightforward to identify overdue items and follow up with the relevant student.

Technical Description:

This screen is powered by an admin endpoint that queries the `borrow_records` table joined with both students and books, with optional WHERE clauses applied based on the filters selected in the React.js interface, and an overdue flag computed by comparing `due_date` to the current date.

10. Authentication and Role-Based Access Control Feature

User Perspective:

Whether logging in as a student or as an administrator, the interface presented afterwards differs automatically: students see catalogue and borrowing tools, while administrators see the management dashboard, without either group being able to access the other's restricted pages.

Technical Description:

Each user's role, student or admin, is stored in the database and embedded as a claim inside the JWT issued at login. Express.js middleware decodes and verifies this token on every protected route, rejecting requests with an invalid token and returning a 403 Forbidden response if a student attempts to reach an admin-only endpoint.

11. API-Driven Frontend-Backend Communication Feature

User Perspective:

To the student or librarian, the application feels instantaneous: search results, borrow confirmations, and admin updates all appear without a full page reload, giving the portal the responsiveness of a modern web application.

Technical Description:

The React.js front end communicates exclusively through a defined set of RESTful JSON endpoints exposed by the Express.js server, using the fetch API or axios to send requests and update component state with the responses. This clean separation means the backend and MySQL database can be modified independently of the user interface, as long as the API contract between them remains stable.

CHAPTER -8 TESTING

8.1 Postman Testing (Frontend/Backend)

The reliability of the Smart Library Database System's application programming interfaces was verified extensively using Postman, an industry-standard tool for designing, sending, and inspecting HTTP requests. Because the React.js front end depends entirely on the Express.js API layer to read and write data in MySQL, confirming that every endpoint behaved correctly in isolation was an essential step before any end-to-end testing could be trusted.

A Postman collection was built covering every major route exposed by the backend, including student registration, login, book search, book details, borrow and return actions, borrowing history, and the full set of administrative routes for managing books, categories, students, and borrowing records. Requests were grouped by resource and tagged by HTTP method, so that GET requests for retrieval, POST requests for creation, PUT requests for updates, and DELETE requests for removal could each be reviewed and re-run independently.

Environment variables were configured within Postman to hold the base API URL and the JWT returned after login, allowing the same collection to be pointed at a local development server or a deployed staging instance simply by switching environments. This meant that the login request could be executed once, its token captured automatically into a variable, and that variable reused across every subsequent protected request without manual copying.

Both valid and deliberately invalid inputs were used to test each endpoint. Registration was tested with a duplicate email to confirm that the API returned a clear conflict response rather than a silent failure; login was tested with an incorrect password to confirm a 401 Unauthorized response; and the borrow endpoint was tested against a book with zero available copies to confirm that the system correctly rejected the request instead of allowing the copy count to go negative.

Postman's built-in test scripts, written in JavaScript within the Tests tab, were used to assert that each response returned the expected HTTP status code and that the JSON body contained the correct fields, such as `student_id`, `book_id`, and `due_date` in a borrow-confirmation response. These assertions turned what would otherwise be manual, visual inspection into a repeatable, automated check that could be re-run after every backend change.

Admin-only routes were tested twice for every case: once with a valid administrator token, to confirm the operation succeeded, and once with a regular student token, to confirm the API correctly returned a 403 Forbidden response. This dual approach validated that role-based access control was enforced consistently across the book, student, and category management endpoints rather than relying solely on the front end to hide restricted options.

Overall, Postman testing gave the development team confidence that the Express.js API layer behaved correctly and predictably before any of it was connected to the React.js interface, catching several early issues with incorrect status codes and missing validation that were far easier to fix at the API level than after the front end had already

been built on top of them.

8.2 Integration Testing

While Postman testing validated each API route on its own, integration testing focused on confirming that the React.js front end, the Express.js backend, and the MySQL database worked together correctly as a single system. The goal was to ensure that a real sequence of user actions, performed through the actual interface rather than through direct API calls, produced the correct end-to-end result in the database.

The core integration scenario tested was the complete borrowing lifecycle: a student registers, logs in, searches for a book, views its details, borrows it, sees it appear in their dashboard and borrowing history, and later returns it. Each step in this sequence was verified both from the interface, by checking that the correct information appeared on screen, and from the database, by directly inspecting the relevant rows in the students, books, and borrow_records tables using MySQL Workbench.

Particular attention was paid to the borrow and return operations, since these are the two points in the system where two related pieces of data, the borrow_records entry and the book's available copy count, must change together. Integration tests deliberately interrupted requests mid-flow, such as by stopping the server between the record insert and the copy-count update during development, to confirm that wrapping both operations in a single database transaction prevented the data from being left in a partially updated, inconsistent state.

Authentication persistence was also tested across the full session lifecycle. After logging in, the JWT stored on the client was used across page navigations, browser refreshes, and multiple API calls to confirm that the student or administrator remained logged in until the token expired or they explicitly logged out, and that an expired or tampered token was rejected on every protected route rather than only on the first one.

Admin-side integration testing confirmed that changes made through the management dashboard, such as adding a new book or updating a category, were immediately reflected in what students saw when they searched the catalogue, since both the student portal and the admin portal read from and write to the same MySQL tables through the same Express.js API.

Together, these integration tests demonstrated that the three layers of the system, presentation, application logic, and data storage, were correctly wired together, and that the guarantees provided at the database level, such as transactional consistency and foreign-key constraints, were actually being exercised by the way the application used them rather than existing only on paper.

8.3 Tools Used for Testing

A small, focused set of testing tools was used throughout development, chosen to cover API validation, automated backend checks, frontend interaction testing, and database verification without introducing more overhead than a project of this scale required.

1. Postman

Used as the primary tool for designing, executing, and organising manual and semi-automated tests against every Express.js endpoint, including authentication, catalogue, borrowing, and admin management routes.

2. Postman Collection Runner

Used to execute an entire saved collection of requests in sequence, allowing the full set of API tests to be re-run automatically after any backend change to catch regressions quickly.

3. Jest

Used to write unit tests for individual backend functions in the Node.js and Express.js codebase, such as input validators and the logic that computes a book's available copy count.

4. React Testing Library

Used to test individual React.js components in isolation, confirming that elements such as the search bar, book card, and borrow button rendered correctly and responded to user interaction as expected.

5. Selenium WebDriver

Used for a smaller set of automated browser tests that simulated a full student journey, from logging in to borrowing a book, to confirm that the interface behaved correctly end to end in an actual browser.

6. MySQL Workbench

Used to inspect the database directly during and after testing, verifying that records in the students, books, categories, and borrow_records tables matched what the application reported through the interface.

7. Git and GitHub

Used for version control of both application code and test scripts, with pull requests reviewed before merging to ensure that new features arrived with corresponding tests already in place.

8. Swagger UI

Used to document the Express.js API endpoints and to allow quick, exploratory testing of a route directly from its documentation page without switching to a separate tool.

9. Google Lighthouse

Used to audit the React.js frontend for performance, accessibility, and best-practice issues, guiding improvements

such as image optimisation and reducing unnecessary re-renders.

8.4 Test Cases and Results

Testing of the Smart Library Database System was carried out across functional, integration, and basic non-functional dimensions to confirm that the application met its design objectives. Functional test cases covered student registration and login, book search and filtering, viewing book details, borrowing and returning books, and reviewing borrowing history, alongside the administrative functions for managing books, categories, students, and borrowing records; each case specified its preconditions, the steps to execute, and the expected outcome, and was run using both Postman for the API layer and manual browser walkthroughs for the interface. Role-based access control was verified by attempting every admin-only action with a student account and confirming that each attempt was correctly rejected. Data-integrity checks confirmed that borrowing a book always decremented its available copy count exactly once and that returning it restored that count, even when multiple borrow and return actions were performed in quick succession. Basic performance observation showed that typical search and borrow operations completed in well under a second on a local development database of a few hundred books and student records, which is well within the expectations for the intended scale of a college or departmental library. A small number of minor issues, mainly around form validation messages and date formatting in the borrowing history view, were identified during testing and corrected before the final build. Across the full set of test cases executed, the system consistently produced the expected results, giving confidence that the application is stable and ready for classroom demonstration and practical use.

CHAPTER -9 DEPLOYMENT

9.1 Steps to Deploy

Deploying the Smart Library Database System and Student Services Portal marks the transition of the project from a locally running development build to an application accessible over the internet to students and library staff. The process was organised into distinct stages covering source control, build preparation, environment setup, and final hosting, ensuring that each part of the stack, the React.js frontend, the Node.js and Express.js backend, and the MySQL database, was deployed in a stable and repeatable way.

The first step was preparing the codebase for release. All development work was committed to a Git repository hosted on GitHub, with the main branch kept in a continuously working state and feature work carried out on separate branches merged in through pull requests. Before any deployment, the full set of Postman and automated tests described in the previous chapter was re-run to confirm that the build being deployed had not introduced any regressions.

With the code confirmed stable, the frontend build process was run using the standard React.js production command, `npm run build`, which compiles and minifies the JavaScript, CSS, and static assets into an optimised bundle suitable for serving directly to browsers rather than through a development server. The backend required no separate compilation step, since Node.js and Express.js run directly from source, but its dependencies were installed cleanly using `npm install` with the `package-lock.json` file to guarantee that the exact same package versions were used in production as during development and testing.

Database preparation followed next. The MySQL schema, comprising the students, books, categories, and borrow_records tables along with their foreign-key relationships, was scripted so that it could be recreated identically on the hosting provider's database instance. A small set of seed data, including a handful of sample books and categories, was inserted so that the deployed application had meaningful content to demonstrate immediately after going live.

Once the database was provisioned, the backend service was deployed to its hosting environment and configured to connect to the newly created MySQL instance using environment variables rather than hard-coded credentials. A short round of manual smoke testing was then performed directly against the deployed API, repeating a handful of the key Postman requests, registration, login, book search, and borrowing, to confirm that the live backend behaved the same way the local development version had.

With the backend verified, the frontend build was deployed and configured to send its API requests to the live backend's URL rather than to localhost. A final end-to-end check was carried out by walking through the complete student journey, and separately the complete admin journey, directly on the deployed site, before the deployment was considered complete and ready for use.

9.2 Environment Configuration

A carefully structured environment configuration underpins the reliable operation of the Smart Library Database System across local development, testing, and production. Because the same codebase runs in all three contexts, every value that differs between them, such as database credentials, API base URLs, and secret keys, is kept out of the source code entirely and supplied instead through environment variables.

On the backend, a `.env` file, excluded from version control through `.gitignore`, defines variables including `DB_HOST`, `DB_USER`, `DB_PASSWORD`, `DB_NAME`, `PORT`, and `JWT_SECRET`. These are read at startup using the `dotenv` package and used to configure the MySQL connection pool and the JWT signing logic, so that switching from a local database to the production database requires changing only the values in this file, not the application code itself.

On the frontend, React.js environment variables, prefixed with `REACT_APP_`, are used to define the base URL of the Express.js API. During local development this points to `http://localhost` together with the backend's development port, while the production build is generated with this variable pointing instead to the deployed backend's public URL, ensuring the compiled bundle never needs to be rebuilt just to change environments.

Dependency management is handled through each project's `package.json` and `package-lock.json` files, one for the React.js frontend and one for the Node.js and Express.js backend, ensuring that both team members and the hosting platform install identical, known-good versions of every library. Scripts defined in `package.json` standardise common tasks, such as `npm start` for local development and `npm run build` for producing the optimised production frontend bundle.

Security-sensitive configuration, particularly the JWT secret and the database password, is never committed to the repository and is instead entered directly into the hosting provider's environment-variable settings, so that these values exist only on the server and in the local developer's untracked `.env` file. This separation between configuration and code is what allows the exact same application to run correctly, and securely, on a laptop during development and on a public server after deployment.

9.3 Hosting of Frontend and Backend

Frontend Hosting

The React.js frontend, once compiled through `npm run build`, produces a set of static HTML, CSS, and JavaScript files that require no server-side processing to display, making them well suited to a static-hosting platform. The build output was deployed using a service such as Vercel or Netlify, both of which integrate directly with the project's GitHub repository so that pushing a new commit to the main branch automatically triggers a fresh build and deployment.

Because the frontend is a single-page application, the hosting configuration includes a rewrite rule directing all routes to `index.html`, allowing React Router to handle navigation between pages such as the search screen, book

details, dashboard, and admin panel entirely on the client side without the hosting platform needing to know about each individual route.

Backend Hosting

The Node.js and Express.js backend, unlike the static frontend, needs to run continuously to accept incoming API requests, so it was deployed to a platform designed for hosting long-running Node.js services, such as Render or Railway. The hosting platform was configured to install dependencies with `npm install` and start the server with `npm start`, restarting the process automatically if it ever crashes.

The MySQL database was hosted using the same platform's managed database offering, or an equivalent managed MySQL service, rather than being run manually on a self-managed server. This provided automatic backups and a stable connection string that the backend's environment variables were configured to use, removing the need for the development team to administer database infrastructure directly.

CI/CD Integration

A lightweight continuous integration workflow was configured through GitHub Actions to run the project's automated tests, including the Jest backend tests and React Testing Library frontend tests, on every push and pull request. Only after these checks passed was a change eligible to be merged into the main branch, which in turn was the branch both the frontend and backend hosting platforms watched for new deployments.

Monitoring and Performance

Basic monitoring was configured through the logging and dashboard features built into the chosen hosting platforms, which surface request logs, error traces, and simple metrics such as response times and uptime for the deployed Express.js service. These logs proved useful during the initial post-deployment period for catching issues such as an incorrectly configured environment variable before they affected real users.

Domain and Networking

The hosting platforms used for both the frontend and backend provide a default subdomain suitable for a college or departmental project, such as `smart-library.vercel.app` for the frontend and a corresponding address for the backend API. Both platforms issue and automatically renew a free SSL certificate, ensuring the deployed site is served over HTTPS by default without any additional configuration.

Scaling and Backup

Given the intended scale of the Smart Library Database System, a departmental or college library serving a defined student population rather than a public, high-traffic service, the chosen hosting tier does not require the complex auto-scaling and load-balancing infrastructure used by large commercial platforms. The managed database's automatic backup schedule provides a straightforward path to recovery in the unlikely event of data loss, which is an appropriate and proportionate level of resilience for a system of this scope.

Post-Deployment Verification and Maintenance

After each deployment, a short verification pass repeats the core student and admin journeys directly on the live site, confirming that registration, login, search, borrowing, returning, and the administrative management screens

all continue to function correctly against the production database. Dependency updates and any necessary security patches to the Node.js runtime or npm packages are applied periodically and re-tested through the same Postman collection described in the testing chapter before being redeployed.

CHAPTER -10 CHALLENGES & LIMITATIONS

10.1 Issues Faced During Development

The development of the **Smart Library Database System** involved several technical and practical challenges. Since

the application combines a frontend interface, backend REST APIs, authentication, database operations, borrowing management, reservation management, fine management, notifications, and administrative functions, maintaining proper communication between all components was an important part of the development process.

One of the first challenges was **database design and relationship management**. The library system contains several interconnected entities such as students, books, categories, borrowing records, reservations, fines, and member history. Designing the database in a way that avoids unnecessary duplication while maintaining correct relationships required careful planning. For example, a borrowing transaction needs to be associated with both a student and a particular book. Similarly, reservation and fine records need to be linked to the appropriate student and transaction.

Another important challenge was maintaining **book availability consistency**. When a student borrows a book, its availability should immediately change so that another student cannot borrow the same copy incorrectly. Similarly, when a book is returned, its availability should be updated. Managing these changes correctly required proper backend validation and database operations.

The project also faced challenges in **frontend and backend integration**. The frontend communicates with the backend using APIs. Each API must receive the correct request parameters, process the request, communicate with the database, and return an appropriate response to the frontend. Any mismatch between frontend requests and backend responses could result in errors or incomplete information being displayed to the user.

The **authentication and authorization system** was another significant area of concern. The application must distinguish between normal students and administrators. Students should be able to access student-oriented services such as searching, borrowing, reservations, and history, while administrators should have access to management functions. Protecting restricted operations from unauthorized users was therefore necessary.

The development team also had to deal with **input validation and error handling**. Users may enter incomplete information, invalid credentials, incorrect book IDs, or other unexpected values. The application therefore needs to validate user inputs before processing them. Meaningful error messages are also required so that users can understand what went wrong.

Testing the application was another challenge. Individual modules may work correctly when tested separately but fail when integrated with other modules. For example, the borrowing module depends on authentication, book availability, student information, and database operations. Therefore, integration testing was required in addition to individual module testing.

The project also considered **temporary network interruptions and offline accessibility**. The documentation specifies offline caching and background synchronization so that previously accessed information can remain available during temporary connectivity problems.

Finally, maintaining the project code and coordinating development activities presented practical challenges. Version control through Git and GitHub and API testing through Postman were included in the project workflow to make development and testing more organized.

Major Development Challenges

The major issues encountered can be summarized as follows:

1. Designing an efficient relational database.
2. Maintaining relationships between students, books, and transactions.
3. Keeping book availability accurate.
4. Connecting the React frontend with backend APIs.
5. Implementing authentication and authorization.

6. Validating user input.
7. Handling API and database errors.
8. Testing multiple modules together.
9. Maintaining consistent data during borrowing and returning.
10. Supporting temporary offline access.
11. Managing notifications and reminders.
12. Maintaining the project using Git and GitHub.
13. Debugging frontend and backend issues.
14. Preparing the application for deployment.

10.2 Solutions Applied

To overcome the challenges encountered during development, several solutions were implemented.

10.2.1 Structured Database Design

A structured MySQL database was designed to store information about students, books, categories, borrowing transactions, reservations, fines, and history. Relationships between the tables were established using suitable identifiers and references.

The database design helped reduce unnecessary duplication and made it easier to retrieve information using SQL queries. Proper table organization also improved the maintainability of the system.

10.2.2 Book Availability Management

Book availability was handled through backend validation. Before allowing a student to borrow a book, the system checks whether the book is currently available.

If the book is available:

- The borrowing transaction is created.
- The student's borrowing information is recorded.
- The book status is updated.
- The transaction becomes part of the student's history.

When the book is returned, the system updates the transaction and makes the book available again.

This approach helps prevent incorrect borrowing operations.

10.2.3 API-Based Communication

The frontend and backend were separated into different layers. The frontend handles the user interface, while the backend manages business logic and database operations.

REST APIs are used to transfer information between these layers. This approach provides a clear structure and makes it easier to maintain individual components.

For example:

Student Interface



React Frontend



REST API



Node.js + Express.js Backend



MySQL Database

The source documentation describes the project as a full-stack application using React.js, Node.js/Express.js, APIs, and MySQL.

10.2.4 Authentication and Authorization

Authentication was implemented to ensure that only registered users can access protected library services.

The system separates users according to their roles. A student can perform operations such as:

- Searching books
- Viewing book details
- Borrowing books
- Returning books
- Reserving books
- Checking fines
- Viewing history

Administrators can perform additional operations such as:

- Adding books
- Updating books
- Removing books
- Managing students
- Managing categories
- Viewing borrowing records
- Managing reservations
- Managing fines

This role-based structure improves system security and prevents unauthorized administrative operations.

10.2.5 Input Validation

Validation was introduced at both frontend and backend levels wherever necessary. Required fields are checked before data is submitted. Invalid or incomplete information is rejected rather than directly stored in the database.

For example, when registering a student, the system can verify that:

- Required fields are provided.
- The email format is valid.
- The password satisfies the required conditions.
- Duplicate accounts are handled appropriately.

Similarly, book information is validated before it is inserted into the database.

10.2.6 Error Handling

Error handling was added to improve reliability and user experience. Instead of allowing application errors to terminate a process unexpectedly, the system returns meaningful responses.

Examples include:

- Invalid login credentials
- Book not available
- Student not found
- Book not found
- Duplicate reservation
- Invalid API request
- Database connection failure

This allows users and administrators to understand problems and take corrective action.

10.2.7 API Testing

Postman was used for testing API endpoints before integrating them completely with the frontend. The project documentation specifically identifies Postman as the API testing tool.

Typical API operations include:

POST /register

POST /login

GET /books

POST /borrow

POST /return

POST /reserve

GET /history

GET /fines

Testing these endpoints independently helped identify backend and database problems before they affected the complete application.

10.2.8 Offline Caching and Synchronization

The project also includes consideration for offline caching. Previously accessed information can remain available during temporary network interruptions, and synchronization can update information after connectivity is restored.

This improves accessibility and provides a better experience when network connectivity is unstable.

10.2.9 Version Control

Git and GitHub were used for version control. This allowed the development team to maintain different versions of the source code, track changes, and recover previous versions when required.

Version control also makes collaborative development easier because developers can work on different components and merge their changes systematically.

10.3 Current Limitations or Known Bugs

Although the Smart Library Database System provides the major functionality required for digital library management, certain limitations remain.

10.3.1 Internet Dependency

The application primarily depends on network connectivity because the frontend communicates with backend APIs and the backend communicates with the database. Although offline caching has been considered, all operations cannot necessarily be performed without an active connection.

For example, viewing previously accessed information may be possible offline, but operations such as borrowing, returning, or modifying records require synchronization with the central database.

10.3.2 Limited Offline Functionality

Offline caching improves accessibility, but it does not provide complete offline functionality. Certain real-time operations require access to the server.

For example, a student should not be allowed to complete a new borrowing transaction entirely offline because the system needs to verify the current availability of the book.

10.3.3 Basic Notification System

The current notification functionality can provide reminders and alerts, but it may not include advanced notification channels such as SMS, push notifications, or personalized notification schedules.

A future version can provide multiple notification channels and allow students to control their notification preferences.

10.3.4 Limited Analytics

The current system focuses mainly on core library operations. Advanced analytics such as predictive borrowing trends, student reading patterns, demand forecasting, and intelligent recommendations are not fully implemented.

10.3.5 No Online Fine Payment

Fine information can be maintained in the database, but an integrated online payment system is considered a future enhancement. The source documentation explicitly identifies online fine payment as a future feature.

10.3.6 No Barcode or QR-Based Book Management

The system can manage books using database records, but automatic barcode or QR-code scanning can make physical library operations faster. Barcode and QR-code scanning are identified as future development possibilities.

10.3.7 Scalability

The current system is primarily designed for a college or departmental library. If the number of users, books, and transactions becomes extremely large, additional optimization may be required.

Possible future improvements include:

- Database indexing
- Query optimization
- Caching
- Load balancing
- Cloud deployment
- Database replication

10.3.8 Dependency on Database Availability

Since important library information is stored in MySQL, database availability is essential for normal operation. A database failure or connection problem can temporarily prevent users from accessing current information.

10.3.9 Limited Recommendation Functionality

The current system does not provide a sophisticated AI-powered recommendation engine. Students may search and filter books, but recommendations based on individual borrowing history and preferences can be introduced later.

CHAPTER – 11

FUTURE ENHANCEMENTS

The Smart Library Database System provides a strong foundation for digital library management. However, the system can be further improved by adding intelligent services, advanced analytics, cloud infrastructure, and better user interaction.

The original project documentation identifies advanced notifications, reports and analytics, cloud deployment, and possible MongoDB integration as future areas.

11.1 Advanced Notification Features

A major future enhancement is the development of an advanced notification system.

The system can automatically notify students about important library activities such as:

- Upcoming due dates
- Overdue books
- Fine generation
- Reservation availability
- Successful borrowing
- Successful book return
- New books added to the library
- Important library announcements
- Reservation expiry
- Account-related notifications

For example, if a student's book is due tomorrow, the system could automatically send:

Reminder: Your borrowed book is due tomorrow. Please return it on time to avoid additional fines.

Notifications can be delivered through multiple channels:

1. In-application notifications
2. Email notifications
3. Push notifications

4. SMS notifications

Students could also be allowed to configure notification preferences.

An advanced notification scheduler could periodically check the database and identify students who require reminders. This would reduce the workload of librarians and improve timely book returns.

11.2 Reports and Analytics

Another important future enhancement is the introduction of a comprehensive reporting and analytics module.

The library administrator could view statistics such as:

- Total number of books
- Total registered students
- Currently borrowed books
- Available books
- Reserved books
- Overdue books
- Total fines
- Most borrowed books
- Most active students
- Popular book categories
- Monthly borrowing statistics
- Monthly return statistics
- Reservation trends

For example, a dashboard could display:

LIBRARY ANALYTICS	

Total Books	: 5,000
Registered Students	: 2,400
Books Borrowed	: 350
Available Books	: 4,450

Overdue Books	: 75
Active Reservations	: 120
Total Pending Fines	: ₹18,500

Graphical charts can also be introduced to make the information easier to understand.

Predictive Analytics

In a more advanced version, historical borrowing data could be analyzed to predict future demand.

For example, if a particular category is frequently borrowed during examination periods, the system could identify this pattern and help librarians purchase additional copies.

This can support:

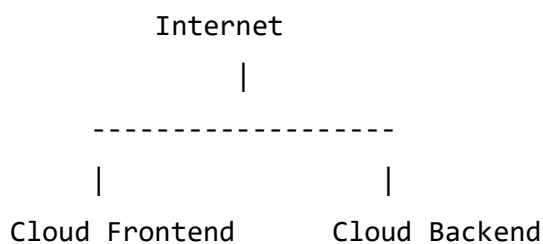
- Inventory planning
- Book purchasing decisions
- Resource allocation
- Demand prediction
- Library management decisions

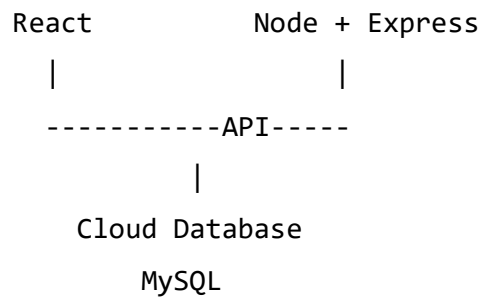
11.3 Cloud Deployment

The application can be deployed completely on cloud infrastructure to improve accessibility, reliability, and scalability.

The source document already discusses deployment using hosted frontend services, long-running Node.js/Express backend services, managed MySQL databases, CI/CD through GitHub Actions, monitoring, HTTPS, and automated backups.

A cloud-based architecture could be:





Advantages include:

Accessibility

Students and administrators can access the system from different devices without requiring the application to be installed locally.

Scalability

Cloud resources can be increased as the number of students and library transactions increases.

Backup

Managed database services can provide automated backup mechanisms.

Security

HTTPS, secure environment variables, authentication, and access controls can be configured for the deployed system.

Continuous Deployment

GitHub Actions or similar CI/CD tools can automatically test and deploy updated versions of the application.

Monitoring

Cloud monitoring tools can track:

- Server uptime
- API response time
- Errors

- Database health
- Application performance

This can make maintenance easier and improve reliability.

11.4 MongoDB Integration if Required

Although the current system uses MySQL for structured library information, MongoDB can be considered for selected use cases if the requirements change.

MySQL is well suited for structured entities such as:

- Students
- Books
- Categories
- Borrowing records
- Reservations
- Fines

MongoDB may be useful when the system needs to store flexible or semi-structured data such as:

- User activity logs
- Search history
- Recommendation data
- Notification preferences
- Analytics events
- Application logs

A future hybrid architecture could therefore use MySQL for core transactional data and MongoDB for flexible analytical or activity-related information.

However, MongoDB should not be introduced unnecessarily. The database technology should depend on actual system requirements, data structure, performance needs, and deployment considerations.

11.5 AI-Based Book Recommendation

One of the most useful future features would be an intelligent book recommendation system.

The system could analyze:

- Previous borrowing history
- Search history
- Book categories
- Frequently viewed books
- Student interests
- Popular books
- Similar users' reading patterns

The system could then provide recommendations such as:

Recommended for You

1. Data Structures and Algorithms
2. Operating System Concepts
3. Database Management Systems
4. Computer Networks

This would make the library system more personalized and intelligent.

11.6 Barcode and QR Code Integration

Barcode or QR-code scanning can improve physical library operations.

Each book can be assigned a unique barcode or QR code. A librarian can scan the code instead of manually entering the book ID.

The system could then automatically:

1. Identify the book.
2. Check availability.
3. Identify the student.
4. Create a borrowing transaction.
5. Update book availability.

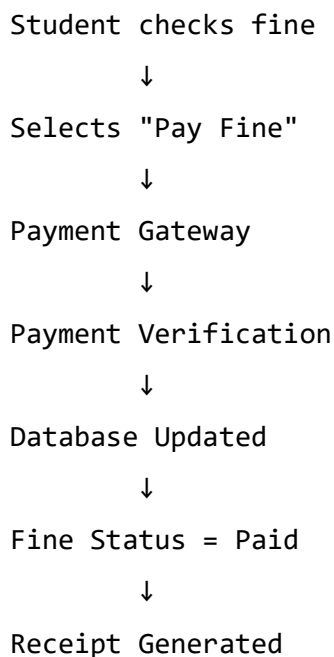
This would reduce manual data entry and improve transaction speed.

Barcode and QR-code scanning are specifically listed among possible future enhancements in the project scope.

11.7 Online Fine Payment

An online payment module can allow students to pay fines directly through the application.

Possible payment workflow:



This would eliminate the need for manual fine collection.

11.8 E-Book and Digital Archive Integration

The library can eventually provide access to digital resources such as:

- E-books
- Research papers
- Academic notes
- Digital journals
- Previous question papers

- Institutional publications

A digital archive would transform the system from a traditional book-management application into a broader academic resource platform.

E-book access and digital archive management are also identified as future possibilities in the project's documented scope.

CHAPTER – 12

CONCLUSION

12.1 Summary of the Project

The **Smart Library Database System** was developed as a web-based solution for simplifying library management and improving student access to library services.

Traditional library processes can involve considerable manual record keeping. Maintaining information about books, students, borrowing, reservations, and fines manually can become difficult as the number of users and transactions increases. The proposed system addresses these problems by providing a centralized digital platform.

The system allows students to:

- Register and log in.
- Search for books.
- Filter books.
- View book details.
- Check book availability.
- Borrow books.
- Return books.
- Reserve unavailable books.
- View fines.
- Check borrowing history.

The administrative section provides librarians with centralized management capabilities for:

- Students
- Books
- Categories
- Borrowing records
- Reservations
- Fines
- Library information

The documented project scope includes registration and authentication, book management, borrowing and returning, reservations, fine management, search and reports, member history, administrative management, REST API communication, MySQL database management, offline caching, synchronization, notifications, Postman API testing, and Git/GitHub version control.

The project follows a layered architecture in which the frontend provides the user interface, the backend manages application logic and APIs, and the database stores the application's persistent information.

12.2 What Was Achieved

Several important objectives were achieved through the development of the Smart Library Database System.

1. Centralized Library Management

The system provides a single platform for managing important library activities. This reduces dependency on separate manual records and makes information easier to access.

2. Student Registration and Authentication

Students can create accounts and log in to access library services. Authentication provides a basic security layer for protecting user information.

3. Book Management

Administrators can maintain book information including title, author, category, and availability.

Books can be:

- Added
- Updated
- Deleted
- Searched
- Filtered
- Organized by category

4. Borrowing and Returning

Students can borrow available books and return them through the system. Borrowing records help maintain an accurate history of library transactions.

5. Reservation Management

Students can reserve books that are currently unavailable. This provides a more convenient method for managing high-demand books.

6. Fine Management

The system maintains fine-related information and allows administrators to track fines associated with library activities.

7. Search and Reports

Students can search and filter books, while administrators can retrieve useful information from the database through queries and reports.

8. Member History

The system maintains historical information about borrowed and returned books. This makes it easier for students and administrators to review previous library activity.

9. Administrative Control

The administrative module provides centralized control over major library operations.

10. API Communication

The application demonstrates communication between frontend and backend through REST APIs.

11. Database Management

MySQL provides structured storage for library data and transactions.

12. Testing

API testing, database testing, integration testing, and authentication testing were included in the project's testing approach.

Overall, the project demonstrates practical application of **full-stack web development, database management, authentication, CRUD operations, API communication, and system integration.**

12.3 Skills Learned During Development

The development of the Smart Library Database System provided practical knowledge in several areas of software development.

Web Development

The project improved understanding of frontend development and the creation of responsive web interfaces.

The development process involved concepts such as:

- HTML
- CSS
- JavaScript
- React.js
- Components
- Forms
- Navigation
- API communication

Backend Development

The project provided practical experience in backend application development using Node.js and Express.js.

Important concepts included:

- REST APIs
- Routes
- Request and response handling
- Middleware
- Validation
- Error handling
- Authentication
- Database connectivity

Database Management

Working with MySQL provided experience in:

- Database creation
- Table design
- Relationships
- Primary keys
- Foreign keys
- SQL queries
- Insert operations
- Update operations
- Delete operations
- Select operations
- Data retrieval

Authentication and Security

The project introduced important concepts related to authentication and authorization, including controlling access to different parts of the application.

API Testing

Postman helped in understanding how APIs can be tested independently before frontend integration.

Debugging

The development process provided practical experience in identifying and resolving:

- Frontend errors
- Backend errors
- API errors
- Database errors
- Authentication problems
- Integration problems

Version Control

Git and GitHub provided experience in maintaining source code versions, tracking changes, and organizing development work.

Software Documentation

The project also improved technical documentation skills, including:

- Requirement documentation
- System design
- Architecture descriptions
- Testing documentation
- Deployment documentation
- References
- Appendices

Teamwork and Problem Solving

The project required planning, implementation, testing, debugging, and documentation. These activities improved problem-solving ability, coordination, communication, and project-management skills.

Overall Conclusion

The Smart Library Database System demonstrates how software technology can be applied to solve practical problems in educational institutions. By digitizing important library operations, the system can reduce manual effort, improve data organization, provide faster access to information, and improve the overall experience of students and librarians.

The current implementation provides a strong foundation for future improvements such as AI recommendations, online payments, barcode scanning, e-book access, advanced analytics, and cloud deployment.

CHAPTER – 13

REFERENCES

The following books, documentation websites, tutorials, and technical resources can be included as references for the Smart Library Database project. The original documentation contains several software-development references; unrelated tourism-specific references should not be included for this library

project.

13.1 Books

1. **Pressman, R. S., & Maxim, B. R.**

Software Engineering: A Practitioner's Approach, 8th Edition, McGraw-Hill Education, 2019.

2. **Sommerville, I.**

Software Engineering, 10th Edition, Pearson Education, 2016.

3. **Duckett, J.**

HTML & CSS: Design and Build Websites, Wiley, 2014.

4. **Duckett, J.**

JavaScript and JQuery: Interactive Front-End Web Development, Wiley, 2015.

5. **Freeman, E., & Freeman, E.**

Head First HTML and CSS, O'Reilly Media.

These books were also listed in the source documentation as software-development and web-development references.

13.2 Online Tutorials and Learning Resources

W3Schools

Used as a learning resource for HTML, CSS, JavaScript, SQL, and web-development concepts.

[W3Schools](https://www.w3schools.com/)

MDN Web Docs

Used for web standards and JavaScript, HTML, CSS, and browser-related documentation.

[MDN Web Docs](https://developer.mozilla.org/en-US/)

FreeCodeCamp

Used as an additional learning resource for web development and programming concepts.

[freeCodeCamp](https://www.freecodecamp.org/)

GeeksforGeeks

Used for programming, data structures, algorithms, SQL, and web-development reference material.

[GeeksforGeeks](#)

Tutorialspoint

Used as a supplementary resource for programming and web-development concepts.

[TutorialsPoint](#)

13.3 Technology Documentation

React.js Documentation

Used for frontend development, components, state management, and application structure.

[React Documentation](#)

Node.js Documentation

Used for backend runtime concepts and Node.js development.

[Node.js Documentation](#)

Express.js Documentation

Used for REST API development, routing, middleware, and server-side application development.

[Express.js Documentation](#)

MySQL Documentation

Used for database creation, SQL queries, relationships, and database management.

[MySQL Documentation](#)

Postman Documentation

Used for REST API testing and verification.

[Postman Documentation](#)

Git Documentation

Used for version control and source-code management.

[Git Documentation](#)

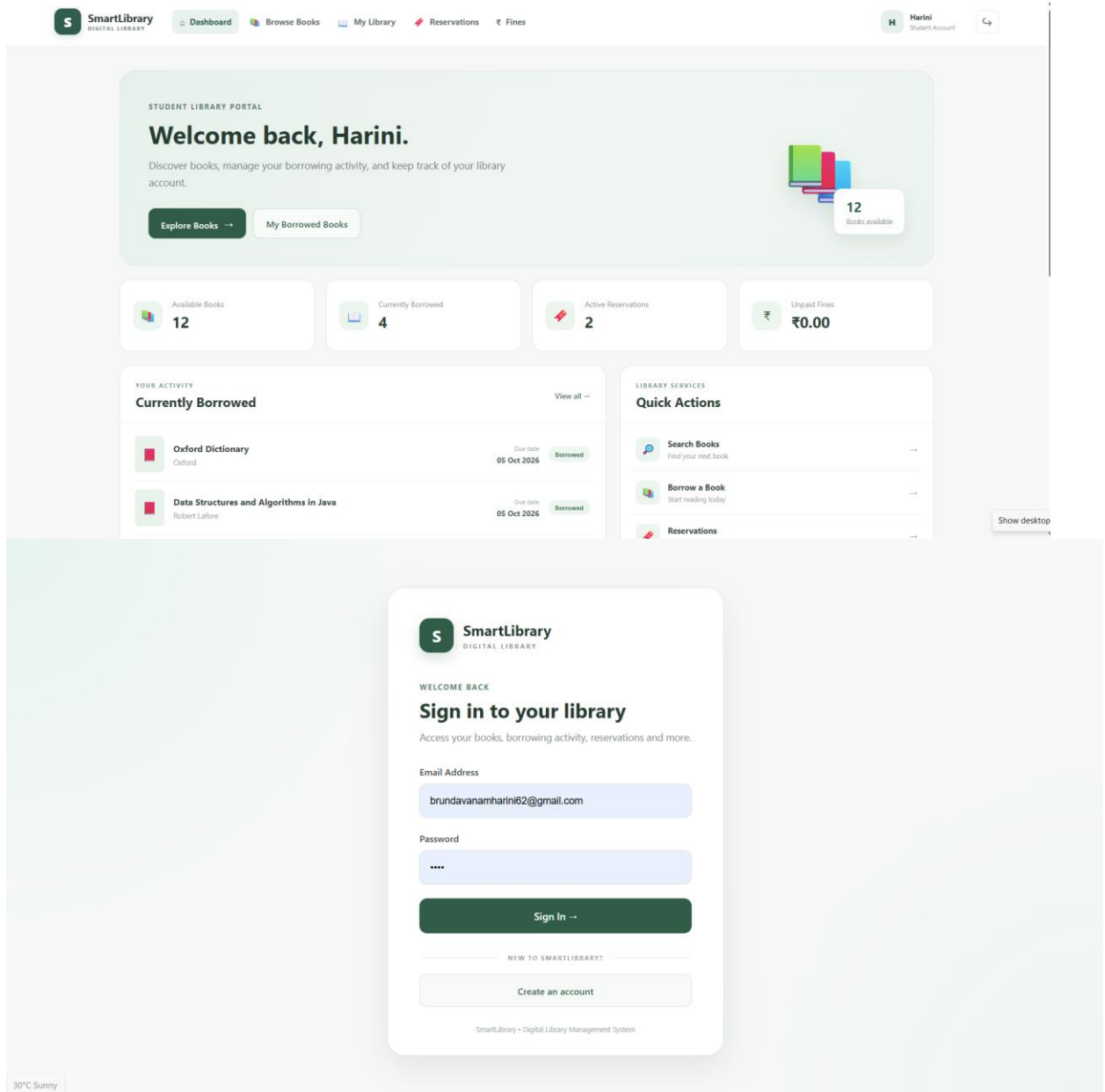
GitHub Documentation

Used for repository management and collaborative development.

[GitHub Documentation](#)

CHAPTER -14 APPENDICES

Screenshots of the app



Sample code snippets

The image displays two sample code snippets from a web application, likely a library management system, shown in a VS Code editor interface.

Snippet 1: App.jsx

```
1 import { useEffect } from "react";
2 import axios from "axios";
3 import { BrowserRouter, Routes, Route } from "react-router-dom";
4 import "./App.css";
5 import Login from "../pages/Login";
6 import Register from "../pages/Register";
7 import StudentDashboard from "../pages/StudentDashboard";
8 import BookSearch from "../pages/BookSearch";
9 import BorrowBook from "../pages/BorrowBook";
10 import MyBorrowedBooks from "../pages/MyBorrowedBooks";
11 import MyReservations from "../pages/MyReservations";
12 import ReserveBook from "../pages/ReserveBook";
13 import MyFines from "../pages/MyFines";
14 import ReturnBook from "../pages/ReturnBook";
15 import BookDetails from "../pages/BookDetails";
16 import Account from "../pages/Account";
17 import AdminDashboard from "../pages/AdminDashboard";
18 import AdminBooks from "../pages/AdminBooks";
19 import AdminAddBook from "../pages/AdminAddBook";
20 import AdminEditBook from "../pages/AdminEditBook";
21 import AdminStudents from "../pages/AdminStudents";
22 import AdminBorrowing from "../pages/AdminBorrowing";
23 import AdminReservations from "../pages/AdminReservations";
24 import AdminFines from "../pages/AdminFines";
25 import AdminReports from "../pages/AdminReports";
26
27 function App() {
28   useEffect(() => {
29     // ...
30   });
31 }
32
33 export default App;
```

Snippet 2: NavBar.jsx

```
1 import { useState } from "react";
2 import { NavLink, useNavigate } from "react-router-dom";
3 import axios from "axios";
4
5 function NavBar() {
6   const navigate = useNavigate();
7   const [menuOpen, setMenuOpen] = useState(false);
8
9   const user = JSON.parse(localStorage.getItem("user"));
10
11   const handleLogout = () => {
12     localStorage.removeItem("user");
13     localStorage.removeItem("token");
14
15     delete axios.defaults.headers.common["Authorization"];
16
17     navigate("/login");
18   };
19
20   const closeMenu = () => {
21     setMenuOpen(false);
22   };
23
24   return (
25     <header className="student-navbar">
26       <div className="student-navbar-inner">
27         <div>
28           <img alt="Logo" />
29           <span>SmartLibrary</span>
30         </div>
31         <div>
32           <NavLink to="/login">Login</NavLink>
33           <NavLink to="/register">Register</NavLink>
34           <NavLink to="/student-dashboard">Student Dashboard</NavLink>
35           <NavLink to="/book-search">Book Search</NavLink>
36           <NavLink to="/borrow-book">Borrow Book</NavLink>
37           <NavLink to="/my-borrowed-books">My Borrowed Books</NavLink>
38           <NavLink to="/my-reservations">My Reservations</NavLink>
39           <NavLink to="/reserve-book">Reserve Book</NavLink>
40           <NavLink to="/return-book">Return Book</NavLink>
41           <NavLink to="/my-fines">My Fines</NavLink>
42           <NavLink to="/account">Account</NavLink>
43           <NavLink to="/admin-dashboard">Admin Dashboard</NavLink>
44           <NavLink to="/admin-books">Admin Books</NavLink>
45           <NavLink to="/admin-add-book">Admin Add Book</NavLink>
46           <NavLink to="/admin-edit-book">Admin Edit Book</NavLink>
47           <NavLink to="/admin-students">Admin Students</NavLink>
48           <NavLink to="/admin-borrowing">Admin Borrowing</NavLink>
49           <NavLink to="/admin-reservations">Admin Reservations</NavLink>
50           <NavLink to="/admin-fines">Admin Fines</NavLink>
51           <NavLink to="/admin-reports">Admin Reports</NavLink>
52         </div>
53       </div>
54     </header>
55   );
56 }
```

Both snippets include a terminal window showing the command to run the application:

```
C:\Users\HARINI\OneDrive\Documents\SmartLibrary>cd frontend
C:\Users\HARINI\OneDrive\Documents\SmartLibrary\frontend>npm run dev
> frontend@0.0.0 dev
> vite
```

Installation/setup instructions

The Smart Library Management System is relatively simple to set up as it uses a React-based frontend, a Node.js and Express.js backend, and a MySQL database. The project is divided into separate frontend and backend folders, making the application easy to install, configure, and run on a local computer.

First, download or copy the complete **SmartLibrary** project folder to the local computer. Open the project in a text editor such as **Visual Studio Code**. The project contains two main folders, **frontend** and **backend**, along with their respective source files and configuration files.

Next, open a terminal inside the backend folder and install the required dependencies using `npm install`. The backend uses Node.js, Express.js, MySQL2, CORS, bcrypt, dotenv, and JSON Web Token packages. The MySQL database should also be created using **MySQL Workbench** with the required tables such as users, books, categories, borrowing, reservations, and fines. The database connection and authentication configuration are provided through the backend environment configuration.

After completing the backend setup, start the backend server using `npm run dev`. The server runs locally and connects the application with the MySQL database. A separate terminal can then be opened inside the frontend folder, where the required dependencies are installed and the React application is started using `npm run dev`. The frontend will open through the local Vite development server in a modern web browser such as Google Chrome, Microsoft Edge, or Firefox.

Once both the frontend and backend are running, the Smart Library Management System can be accessed through the browser. The application requires the backend server and MySQL database to be active so that features such as login, registration, book borrowing, reservations, returns, and fine management can function correctly.

User Manual / Guide

When the Smart Library Management System is opened, users are presented with a clean login interface where they can either log in using their existing credentials or register as a new student. After successful login, students are directed to the student dashboard, where they can access the main features of the library system.

Students can browse and search for books, view detailed information about each book, borrow available books, and reserve books when required. The **My Borrowed Books** section allows students to track their active and returned books along with borrowing and due dates. The system also provides reminders when a book is approaching its due date and allows eligible borrowing periods to be extended. Students can view their reservations and check any unpaid or previous fines through the respective sections.

The platform also provides a separate **Administrator Portal**. Administrators can manage books and categories, view registered students, monitor borrowing and reservation records, manage fines, update due dates, and view library reports. Authentication and role-based access ensure that students and administrators are provided with the appropriate features according to their roles.

Overall, the Smart Library Management System provides a simple and organized digital environment for managing common library activities. It demonstrates how book management, borrowing, reservations, returns, due-date tracking, authentication, and fine management can be integrated into a single web-based application.

Geo Tag photos with guide

Review forms with guide signatures

